Certifying Deep Learning Classifier Predictions with High Confidence

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Abstract

A classifier's test-set error gives no high-confidence guarantee about its true risk. PAC-Bayes theory provides such guarantees, called risk certificates, for stochastic predictors, and the PAC-Bayes-with-Backprop (PBB) framework of Pérez-Ortiz et al. [20] makes them numerically non-vacuous for deep networks by training the weight distribution to minimise the bound directly. A certificate is only valid, however, if every step of its computation is correct. This dissertation re-implements that computation from scratch, reproduces four MNIST cells of the reference under a reduced-compute protocol, and extends the same pipeline to Fashion-MNIST and CIFAR-10.

We found and corrected five issues in the reference implementation, each under a unit test: three change the reported certificate (the bound-set sample count, the batched Monte-Carlo aggregation and the joint confidence accounting), one affects reproducibility (seed control), and one decides whether the learnt prior is admissible at all, since a class-stratified partition makes the prior depend on the very labels the bound is evaluated on. We additionally correct for a selection of our own: the prior scale was settled by comparing certificates on the bound set, so every certificate carries a union over a grid of candidate prior scales, under a stated assumption we carry as a threat rather than a settled matter.

Over five seeds the reproduction matches the published stochastic test errors to within 0.0033 on all four target cells, while our certificates are looser by $+0.0214$ to $+0.0423$; re-drawing at the reference's full Monte-Carlo budget recovers a little over half of that gap on the checkpoint examined, and the remainder is left unattributed. The learnt-prior MNIST certificates are non-vacuous (f_{classic} 0.0455, f_{quad} 0.0580) and loosen to 0.1746 on Fashion-MNIST and 0.4277 on CIFAR-10, where the 13-layer network is tightest (0.3881) and the 15-layer one trains only three seeds in five. A data-dependent prior tightens the certificate in all three paired controls, though not by the same mechanism in each, and the accuracy-oriented Bayes-by-Backprop baseline attains the best test error and the loosest certificate: tightness tracks not accuracy but how much KL an objective spends to reach it. Decomposing the certified risk by class shows it is far from evenly spread, one Fashion-MNIST class carrying 0.293 against a dataset-wide 0.108. We certify only the Gibbs classifier, and release an artefact in which every reported number is regenerated from a results registry. The main contribution is the corrected, tested pipeline, since the value of any certificate depends on the sample counts, confidence accounting and Monte-Carlo aggregation that produce it.

Declaration of originality

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Acknowledgments

I thank my supervisor, Dr. Omar Rivasplata, for detailed feedback on the correctness of the risk certificates and on the reproducibility of the experimental pipeline. This work builds on the open-source PAC-Bayes-with-Backprop implementation of Pérez-Ortiz et al. (2021); the public MNIST, Fashion-MNIST and CIFAR-10 benchmarks were used throughout. Generative AI tools were used in the course of the project: OpenAI Codex to review the experiment code before the experiments were run, and GPT-5.6, through the ChatGPT interface, to review this report and to edit its language. Their use is declared in full in the accompanying declaration of generative AI use.

1 Introduction

1.1 Motivation

When a neural network is reported to reach, say, 98% accuracy on a benchmark, that figure is an estimate from one finite test set. It carries no stated confidence, it varies from one test split to another, and it can be quietly inflated when the same test set is consulted repeatedly during model selection. For a deployed classifier we often want something stronger: "with probability at least 99%, the expected error of this model is below r". A bound of this kind is a *risk certificate*, computed from the training data and the learning algorithm rather than from a held-out test set.

PAC-Bayes theory [16, 22, 2] is the main route to such guarantees, for *stochastic* predictors whose weights are drawn from a distribution Q learnt from data. For a long time these bounds were of theoretical interest only: applied to a modern over-parameterised network they returned values above one, which certify nothing. That changed when the bound was used not merely to analyse a trained model but to *train* it. PAC-Bayes-with-Backprop (PBB) [20] optimises the weight distribution by minimising a PAC-Bayes bound directly, obtaining certificates that are numerically meaningful and competitive in accuracy. This project reproduces that framework, extends it to two further datasets, and examines how trustworthy the resulting certificates are once the computation behind them is scrutinised.

1.2 Why certificate reproduction is hard

A test-error number is forgiving: a small mistake in computing it usually produces a slightly wrong but still plausible figure. A certificate is not. Its validity rests on a chain of conditions that must all hold at once. The empirical risk inside the bound has to be measured on data the prior never saw, and the sample size dividing the complexity term has to be the size of that held-out set rather than of the whole dataset. The risk of a stochastic predictor has no closed form, so it is estimated by Monte-Carlo sampling, which introduces a second failure probability that must be combined with the PAC-Bayes one. And the object certified is the stochastic (Gibbs) classifier, not the deterministic network a practitioner deploys. Handle any of these wrongly and the pipeline still produces a number that looks like a certificate, but it no longer holds.

We therefore re-implement the pipeline from first principles alongside reproducing and extending PBB, and test each condition explicitly. That surfaced several points at which the reference implementation departs from them; we correct those and recompute every certificate accordingly.

1.3 Subject area and key related work

Non-vacuous numerical bounds go back to Langford and Caruana [13], for small networks only. The modern line begins with Dziugaite and Roy [4]: a deterministic network has a point-mass weight distribution, for which the PAC-Bayes complexity term is infinite, so by spreading the weights into a Gaussian and optimising the spread they turned an infinite penalty into a finite one and reported the first non-vacuous bound (around 0.21) on MNIST. It was loose, but it established that the approach was numerically viable.

Two factors separate a loose bound from a tight one, and both matter here. The first is keeping the posterior Q close to the prior P, since the bound penalises their KL divergence, and a prior fixed before seeing any data sits far from any well-trained posterior. Dziugaite and Roy [5] addressed this by making the prior depend on the data through differential privacy, and Rivasplata et al. [21] analysed instance-dependent priors more generally; both show that moving the prior towards the data shrinks the KL substantially. The second is to optimise the bound itself. Pérez-Ortiz et al. [20], the work this project reproduces, trains Q by minimising one of three surrogate PAC-Bayes objectives (a McAllester/Pinsker form f_{classic}, the Catoni-style bound of Thiemann et al. [23] f_λ, and a quadratic relaxation f_{quad}), combined with a prior learnt on a held-out portion of the training data, and reports 0–1 certificates around 0.02–0.03 on MNIST, an order of magnitude tighter than Dziugaite and Roy [4], at good accuracy. Later work develops the self-certified and learnt-prior settings [19, 18], and García-Pérez, Parrado-Hernández, and Shawe-Taylor [7] tightens the underlying inequalities.

Which predictor the bound controls matters throughout what follows: it is a statement about the Gibbs classifier, which resamples weights for each prediction. Transferring it to the deterministic majority vote is possible in principle through the C-bound [11, 8], which relates the majority-vote risk to the mean and variance of the Gibbs risk, but that is a different and generally looser statement. The baseline we compare against, Bayes-by-Backprop [1], learns a Gaussian posterior by maximising a variational objective rather than a risk bound, and so achieves strong accuracy without a tight certificate. More broadly, deep ensembles [12], MC-dropout [6] and post-hoc calibration [9] quantify uncertainty empirically without a distribution-free guarantee on the risk, while randomised smoothing [3] certifies a different, adversarial quantity.

1.4 Objectives

The project brief is to reproduce the PBB experiments on MNIST, extend them to Fashion-MNIST, and analyse CIFAR-10. Within that brief the technical work pursues four questions. The first is one of *reproduction fidelity*: under an explicitly reduced compute budget, how closely does a corrected, independent re-implementation match the reference's MNIST results when the same metrics are compared? To make that decidable rather than a matter of impression, targets and tolerance are fixed in advance: four MNIST cells of the reference's Table 1 — FCN with a data-free prior under f_{quad}, FCN with a learnt prior under f_{quad} and under f_λ, and CNN with a learnt prior under f_{quad} — each counting as reproduced when our mean stochastic test error is within one percentage point of the published value. We set *no* tolerance on the certificate, because one computed at $m = 2{,}000$ is not the same quantity as one computed at $m = 150{,}000$; instead we report the gap per cell and measure how much of it the Monte-Carlo budget explains. The second concerns the *anatomy* of the certificate, namely how the empirical Gibbs risk, the KL term and the bound-set size n_{b} each contribute to the final value across the three datasets, and how a learnt prior shifts that balance. Building on this, the third asks how *sensitive* the certificate is to the choices around it: the prior family, the amount of training data, the Monte-Carlo budget, the random seed, and the sample-count convention that the audit corrects. The last isolates a single notion of difficulty: with the architecture, sample size and compute held fixed, does injected label noise move the empirical-risk, KL and certificate terms together, so that a difficulty effect can be separated from the confounds of a cross-dataset comparison?

The dissertation makes five contributions. It (i) reproduces the principal PBB MNIST cells under a documented, reduced-compute protocol and compares them with the published numbers on identically-defined metrics; (ii) extends the same pipeline to Fashion-MNIST and CIFAR-10, adding the learnt-versus-random prior contrast on every dataset, a CIFAR-10 random-prior control the original study did not report, and its 13- and 15-layer CIFAR-10 architectures, at the deeper of which the fixed configuration begins to fail to train; (iii) identifies, in the public reference *implementation* and not in the theory of Pérez-Ortiz et al. [20], whose inequalities we use unchanged, five issues in the certificate pipeline, and corrects each under a unit test: three change the certificate value (the bound-set sample count, the batched Monte-Carlo aggregation, the joint confidence accounting), one affects reproducibility (seed control) and one affects whether the learnt-prior construction meets the independence condition at all (a label-independent partition). It also pays for a step of its own, since the prior scale was chosen by comparing certificates on the bound set, with a union over a grid of candidate prior scales, and states precisely what that correction does and does not settle; (iv) decomposes certificate behaviour into its empirical-risk, KL, sample-size and Monte-Carlo components, by class, and under a controlled label-noise perturbation; and (v) releases a reproducible artefact in which every results figure and reported number is generated from a results registry with per-run provenance and an inclusion/exclusion record.

1.5 Report structure

Section 2 sets out the theory and the assumptions it rests on, Section 3 the experimental design and the corrected pipeline, and Section 4 the reproduction, the certificate anatomy, the sensitivity analyses and the controlled-difficulty study. Section 5 reflects on validity and limitations, and Section 6 concludes.

2 Theory and Methodology

2.1 Setup and assumptions

We consider multiclass classification with inputs $x \in \mathcal{X}$ and labels $y \in \{1, \ldots, K\}$. A training sample $S = \{(x_i, y_i)\}_{i=1}^n$ is drawn *i.i.d.* from an unknown distribution $\mathcal{D}$. A stochastic predictor is a distribution Q over weight vectors w; drawing $w \sim Q$ and predicting $f_w(x)$ defines the *Gibbs classifier*. For a loss ℓ taking values in $[0, 1]$ the Gibbs risk and its empirical counterpart on a set T are

$$R(Q) = \mathbb{E}_{w \sim Q}\, \mathbb{E}_{(x,y)\sim\mathcal{D}}\, \ell(f_w(x), y), \qquad \hat{R}_T(Q) = \mathbb{E}_{w\sim Q}\, \frac{1}{|T|} \sum_{(x,y)\in T} \ell(f_w(x), y). \tag{1}$$

A certificate bounds the population risk using the empirical quantity. PAC-Bayes bounds require (i) i.i.d. data, (ii) a loss bounded in $[0, 1]$, and (iii) a prior P chosen *independently of the sample used to evaluate the empirical risk in the bound*. The 0–1 loss is bounded directly; the cross-entropy is not, so wherever it enters a bound we use the standard bounded surrogate of Dziugaite and Roy [4], clamping log-probabilities at $\log p_{\mathsf{min}}$ and rescaling by $1/\log(1/p_{\mathsf{min}})$, with $p_{\mathsf{min}} = 10^{-5}$.

2.2 The PAC-Bayes-kl certificate

All formal certificates in this dissertation are instances of the PAC-Bayes-kl inequality [15, 22], the tightest of the classical PAC-Bayes statements. Writing $\mathsf{kl}(q\|p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$ for the binary KL, it states that for any prior P independent of the bound set S_{b} of size n_{b}, simultaneously for all posteriors Q, with probability at least $1 - \delta$ over the draw of S_{b},

$$\mathsf{kl}\big(\hat{R}_{S_{\mathsf{b}}}(Q) \,\big\|\, R(Q)\big) \;\le\; \frac{\mathsf{KL}(Q\|P) + \log \frac{2\sqrt{n_{\mathsf{b}}}}{\delta}}{n_{\mathsf{b}}}. \tag{2}$$

The certificate is obtained by inverting (2): with $\mathsf{kl}^{-1}(u; c) = \sup\{v \in [u, 1) : \mathsf{kl}(u\|v) \le c\}$,

$$R(Q) \;\le\; \mathsf{kl}^{-1}\Big(\hat{R}_{S_{\mathsf{b}}}(Q)\,;\, \tfrac{1}{n_{\mathsf{b}}}\big(\mathsf{KL}(Q\|P) + \log \tfrac{2\sqrt{n_{\mathsf{b}}}}{\delta}\big)\Big), \tag{3}$$

which has no closed form and is evaluated by bisection. Two quantities govern it: the empirical Gibbs risk $\hat{R}_{S_\mathsf{b}}(Q)$ and the complexity term $\mathsf{KL}(Q\|P)/n_\mathsf{b}$. The sample size in (2) is n_b, the bound set on which the empirical risk is measured, not the full dataset; Section 2.5 returns to why that matters.

2.3 What the certificate covers: the Gibbs classifier

The object bounded by (3) is the Gibbs classifier, which freshly samples $w \sim Q$ for each prediction — rarely what a practitioner deploys, since sampling per example is wasteful and makes the prediction random. We therefore report three predictors: the *stochastic* (Gibbs) one, which the certificate controls; the *posterior-mean* predictor $\mathbb{E}_Q[w]$, the natural deterministic deployment; and the *finite ensemble* averaging the softmax outputs of L draws. Reporting all three shows how far the certified quantity sits from what would be deployed.

The posterior-mean and ensemble errors are functions of Q but are *not* controlled by (2): a low ordering on a finite test set does not transfer the Gibbs certificate to a deterministic predictor. A formal guarantee for the majority vote is obtainable in principle from the Gibbs risk and its variance via the C-bound [11, 8], but it is a different, generally looser statement that we do not compute here.

2.4 From finite Monte-Carlo to a certificate: two failure probabilities

The empirical Gibbs risk $\hat{R}_{S_\mathsf{b}}(Q)$ in (3) is an expectation over $w \sim Q$ and cannot be computed exactly; it is estimated from m independent Monte-Carlo replicates. This introduces a *second* source of randomness with its own failure probability, which must be accounted for separately from the PAC-Bayes one, so we split the confidence budget in two and combine the parts by a union bound. Let $\bar{R}$ be the mean 0–1 error over the m replicates and the n_b bound examples. A replicate scores one draw $w \sim Q$ on the whole bound set, or, where memory forces that set into several batches, one independent draw per batch. Either way a replicate is an unbiased $[0,1]$-valued estimate of $\hat{R}_{S_\mathsf{b}}(Q)$ and the m of them are i.i.d., which is what (4) needs: the count entering the slack is m, not the number of forward passes. First, inverting the binary KL with the Monte-Carlo slack controls the sampling error with probability $\geq 1-\delta_{\mathsf{MC}}$ [13]:

$$\hat{R}_{S_\mathsf{b}}(Q) \;\leq\; \overline{R}^{+} := \mathsf{kl}^{-1}\Big(\bar{R};\, \tfrac{1}{m}\log\tfrac{2}{\delta_{\mathsf{MC}}}\Big). \tag{4}$$

Second, substituting $\overline{R}^{+}$ for $\hat{R}_{S_\mathsf{b}}(Q)$ in (3) at confidence δ_{PAC} yields the population certificate. By a union bound over the two events that can fail, the final certificate holds with probability at least

$$1-\delta_{\mathsf{MC}}-\delta_{\mathsf{PAC}}. \tag{5}$$

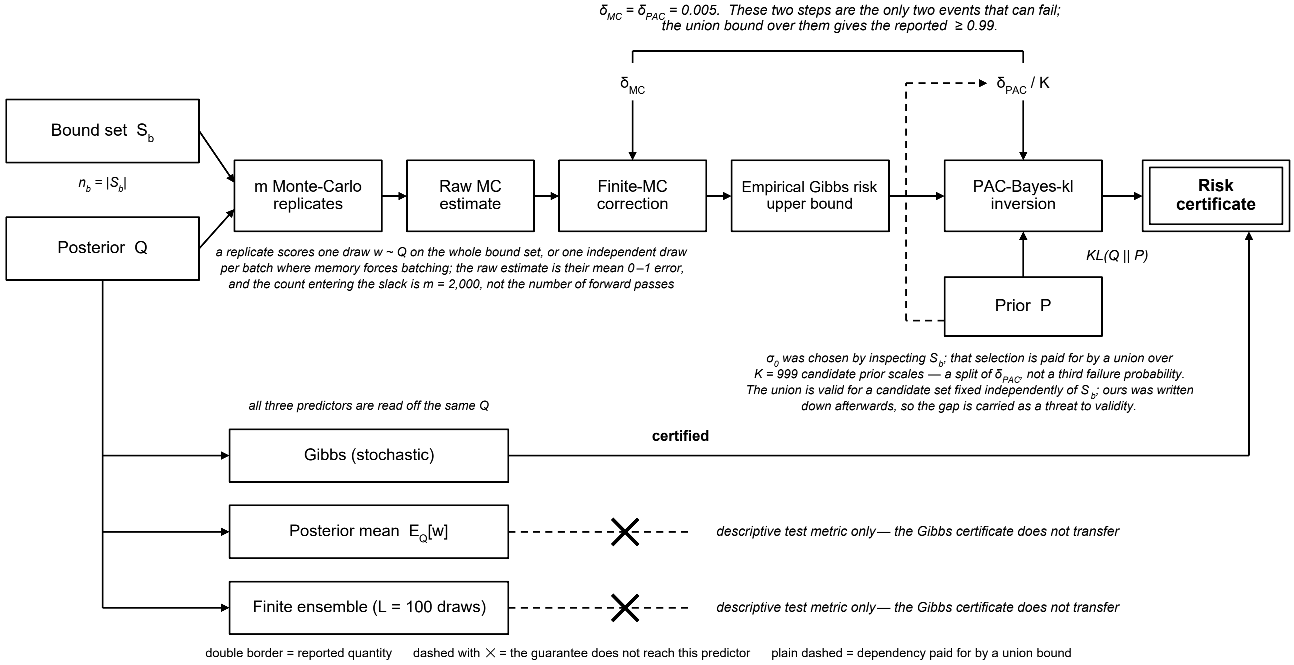


Fig. 1. The certificate computation, end to end. Two distinct corrections stand between the raw Monte-Carlo estimate and the reported number, and the two leader lines mark them as the only two events that can fail: their union is what gives the 0.99. The dashed branch from the prior records that σ_0 was chosen by inspecting S_{b}, which is paid for by splitting δ_{PAC} across the candidate grid rather than by a separate failure probability. Bottom right: all three predictors come from the same Q, but only the Gibbs one is certified, and the crossed lines mark that the guarantee does not transfer to the other two.

It is the sum in (5), not either δ alone, that we report. With $\delta_{\mathsf{MC}} = \delta_{\mathsf{PAC}} = 0.005$ every reported certificate holds with probability at least 0.99; reusing a single $\delta = 0.01$ across both inversions, as the reference does, would double-count one failure probability and overstate the confidence. A third parameter $\delta_{\mathsf{train}} = 0.025$ enters only the training objective (Section 2.6) and never the reported certificate; all three are recorded separately for every run. The headline certificate throughout is the 0–1 one; the cross-entropy certificate is auxiliary, and we do not claim the two hold *simultaneously* at 0.99, which would need a further union split. Figure 1 traces the whole computation.

Choosing the prior spends part of the same budget. The prior scale σ_0 was not fixed in advance: it was settled after comparing certificates computed on S_{b} (Section 3.5). Inequality (2) holds for a prior fixed independently of S_{b}, simultaneously over all posteriors; it does not licence inspecting S_{b} and then picking one prior out of several for free. The repair is one more union bound: applying (2) to each of K candidate priors at δ_{PAC}/K makes the guarantee simultaneous over all of them, so whichever is adopted is covered, at a cost of $\log K$ in the numerator of the slack.

The obvious candidate set is the wrong one. Counting the scales we compared, the three probed and the one adopted, gives $K = 4$; but 0.03 was picked *after* those curves were seen, so that set is itself a function of S_{b} and a union over it covers nothing. We union instead over $\Sigma = \{0.001, 0.002, \ldots, 0.999\}$, every three-decimal scale in the unit interval, so $K = 999$. The price, log 999 nats against log 4, raises the headline certificates by about 0.0005 and the smallest bound set in the study, the 10%-data cell at $n_{\text{b}} = 3{,}000$, by 0.004; no comparison here turns on that. We apply it to every cell, conservatively so for Fashion-MNIST and CIFAR-10, whose bound sets were never consulted in choosing σ_0.

The union is valid for any Σ fixed independently of S_{b}, and then covers the adopted scale however it was picked from Σ. What we cannot show is that our selection was confined to Σ in advance: Σ was written down afterwards, and a procedure returning 0.0325 would not have been covered. The certificates therefore hold under the assumption that only a three-decimal scale could have been produced, which we believe but cannot demonstrate from the artefact. Pre-registration is one way to close that, but not the only one: it would equally suffice to show the candidate set fixed in advance by the configuration space or by the search procedure itself. Ours was not, so we carry it as a threat to validity (Section 5.2) rather than as a solved problem.

The posterior needs no such treatment, since (2) is already simultaneous over all Q: the learning rate, the epoch count and everything else shaping Q may depend on the data freely. That licence covers the PAC-Bayes event only, not the finite-Monte-Carlo step (4), which holds for the Q evaluated rather than simultaneously over candidates. Choosing a checkpoint by comparing finite-m certificates would therefore need its own correction; we do not, computing each certificate once on the final-epoch posterior.

2.5 Data partition: S, S_0 and $S \setminus S_0$

A learnt (data-dependent) prior is built by training on part of the data, which is legitimate only if the bound set is kept independent of the prior. From the full training set we select a subset S, which is the whole training set unless we are studying data scarcity. For a learnt prior we split S into a *prior subset* $S_0 \subset S$ (half of S) used to train the prior, and the disjoint *bound subset* $S_{\text{b}} := S \setminus S_0$ on which the empirical risk entering the certificate is measured. The posterior Q may be trained on all of S; the prior P depends only on S_0 and is therefore independent of S_{b}, as (2) requires. Consequently

$$n_{\text{post}} = |S|, \qquad n_{\text{b}} = |S_{\text{b}}|. \tag{6}$$

The second equality in (6) is the one the audit of Section 3.4 restores: it is the bound subset, not the dataset, that supplies the denominator. For a data-free (random) prior there is no S_0, the prior is independent of all of S, and $n_{\text{b}} = |S|$. On the full MNIST training set with an even split this gives $n_{\text{post}} = 60{,}000$ and $n_{\text{b}} = 30{,}000$ ($50{,}000$ and $25{,}000$ for CIFAR-10); putting $60{,}000$ in the denominator instead, as a naïve reading of the partition suggests, understates the complexity term and returns a

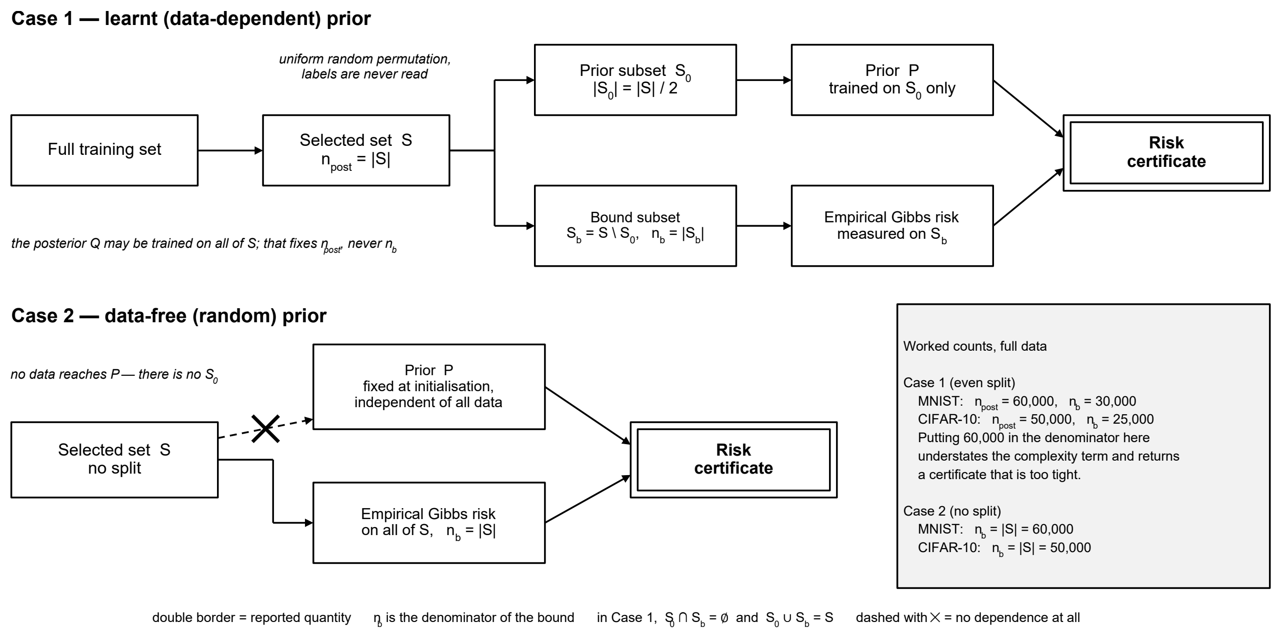


Fig. 2. Data partition and the two prior cases. A uniformly random subset S is drawn from the full training set, and in Case 1 split again into a prior subset S_0 and a bound subset $S_\mathsf{b} = S \setminus S_0$. Both draws are made by permutation alone and never read a label, which is what leaves P admissible; Figure 3 gives the dependence this buys. The certificate's denominator is the size of whichever set the empirical risk is measured on, so it is $n_\mathsf{b} = |S_\mathsf{b}|$ with a learnt prior and $n_\mathsf{b} = |S|$ with a data-free one — the distinction the sample-count correction of Section 3.4 restores.

certificate that is too tight. We therefore derive n_b from explicit bound-subset membership and verify $S_0 \cap S_\mathsf{b} = \varnothing$ and $S_0 \cup S_\mathsf{b} = S$ before every run.

Figure 2 fixes which examples go where; what the bound constrains is which subset may *inform* which quantity (Figure 3).

2.6 Training objectives

The posterior is trained by PAC-Bayes-with-Backprop [20]: the weight distribution Q (a diagonal Gaussian, mean and per-weight scale) is optimised by gradient descent on a differentiable objective derived from a PAC-Bayes bound, through the reparameterisation trick [1]. Whether that objective is itself a bound depends on β, settled below. We compare three bound-derived objectives and one accuracy-oriented baseline, all on $n = n_\mathsf{post}$ training points with bounded empirical cross-entropy $\hat{R}(Q)$ and confidence δ_train: Writing $\kappa = \beta\,\mathsf{KL}(Q\|P) + \log\left(2\sqrt{n}/\delta_\mathsf{train}\right)$ for the complexity numerator shared by the first three:

- f_classic (McAllester/Pinsker): $f = \hat{R}(Q) + \sqrt{\kappa/(2n)}$.

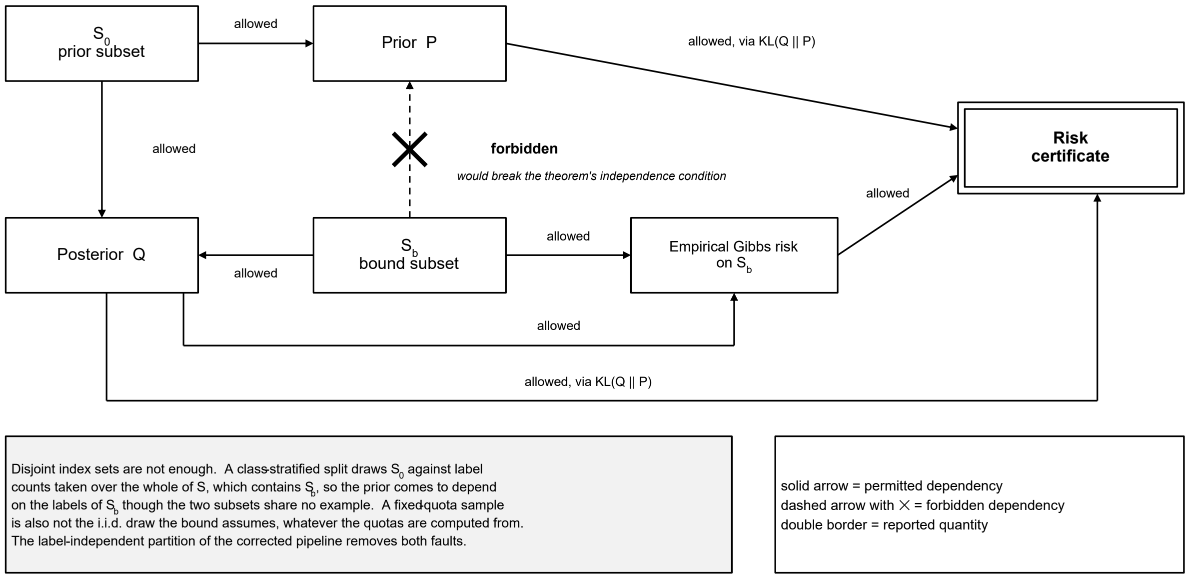


Fig. 3. Permitted and forbidden information flow. The posterior may be informed by every part of S; the prior may be informed only by S_0. The one forbidden edge is $S_\text{b} \to P$, and the note records why disjoint index sets are not by themselves enough to rule it out: a class-stratified split sets S_0's quotas from label counts taken over all of S, so the prior can depend on the bound set's labels without sharing a single example with it. That is the failure the label-independent partition of Section 3.4 removes.

- f_λ [23]: for a learnable $\lambda \in (0, 2)$, optimised by alternating minimisation over Q and λ,

$$f = \frac{\hat{R}(Q)}{1 - \lambda/2} + \frac{\kappa}{n\,\lambda\,(1 - \lambda/2)}.$$

- $f_\textbf{quad}$: the quadratic relaxation of PAC-Bayes-kl [20], which with $\rho = \kappa/(2n)$ is

$$f = \left(\sqrt{\hat{R}(Q) + \rho} + \sqrt{\rho}\right)^2.$$

  It is reported there as the tightest of the three with a data-dependent prior, though in our runs f_classic is tighter (Section 4.2).

- $f_\textbf{bbb}$ (baseline): the Bayes-by-Backprop objective [1], $f = \hat{R}(Q) + \beta\,\text{KL}(Q\|P)/n$. It has the ELBO's form but is not the ELBO: $\hat{R}(Q)$ here is the clamped, rescaled cross-entropy of Section 2.1 rather than a log-likelihood, and $\beta < 1$ down-weights the KL. It trades that KL against the empirical risk with a calibration that does not target the reported certificate, which is what makes it an accuracy-oriented control.

The KL penalty β is carried by all four objectives, not only the baseline, and following the reference's configuration we set $\beta = 0.1$ throughout. This makes each of the first three a down-weighted surrogate rather than the literal bound it is named after, so the value each optimises is not itself a valid certificate. That does not affect what we report, because the objective only shapes Q: the reported certificate is always the *same* PAC-Bayes-kl inversion (3) applied to the resulting Q, evaluated with the unpenalised KL$(Q\|P)$. No ordering of objectives by certificate value is therefore fixed independently of the Q they produce.

2.7 Priors

We study two prior families. A *data-free* (random) prior is a fixed isotropic Gaussian at the initialisation, independent of all data: the most distribution-independent choice, but one that typically leaves a large KL$(Q\|P)$ and a loose certificate. A *data-dependent* (learnt) prior centres P at the weights of a network trained by empirical-risk minimisation on S_0, with dropout to limit prior over-fitting, which moves P close to well-performing posteriors and can sharply reduce the KL. The posterior is initialised at the prior in both cases, and the legitimacy of the learnt one rests on the independence of P from $S \setminus S_0$ (Section 2.5).

3 Experimental Method

3.1 Datasets and preprocessing

We use three standard image-classification benchmarks: **MNIST** [14] (60,000 train / 10,000 test, 28×28 greyscale, 10 classes), **Fashion-MNIST** [24] (same shape, harder classes), and **CIFAR-10** [10] (50,000 / 10,000, 32×32 colour, 10 classes). The only preprocessing is conversion to a tensor and per-channel standardisation with the published training statistics: mean 0.1307 and standard deviation 0.3081 for MNIST, 0.2860 and 0.3530 for Fashion-MNIST, and $(0.4914, 0.4822, 0.4465)$ and $(0.2023, 0.1994, 0.2010)$ for CIFAR-10, applied to train and test alike. We avoid augmentation, resizing and pretraining so that the certificate reflects the PAC-Bayes machinery rather than auxiliary engineering, and treat the cross-dataset comparison as descriptive only, since architecture and compute budget change across datasets (Section 5).

3.2 Architectures

Two families of probabilistic network are used, both carrying a diagonal-Gaussian weight distribution with a mean and a per-weight scale for every parameter, so that a probabilistic network holds twice as many parameters as its deterministic counterpart (Table 1). Every hidden activation is a ReLU, the output is a log-softmax over the ten classes, and no batch normalisation, weight decay or learning-rate schedule is used anywhere. Dropout appears only in the deterministic network that builds a learnt prior, where it limits prior over-fitting to S_0; the certified probabilistic networks have none, since the weight noise already regularises them and a stochastic mask would add a second, uncertified source of randomness to the Monte-Carlo estimate. The prior network shares the architecture of its probabilistic counterpart, so its trained weights can initialise the prior mean directly.

Table 1. Layer specifications of the five network variants used. Convolutions are written as (input channels $\rightarrow$ output channels, kernel, padding), and `MaxPool` is 2×2 with stride 2. Parameter counts are for the deterministic network; the probabilistic counterpart carries twice as many, one mean and one scale per weight.

Network	Layers	Parameters
FCN (MNIST, Fashion-MNIST)	$784 \rightarrow 600 \rightarrow 600 \rightarrow 600 \rightarrow 10$, fully connected	$1{,}198{,}210$
CNN, 4 layers (MNIST, Fashion-MNIST)	Conv($1 \rightarrow 32$, 3×3, pad 0), Conv($32 \rightarrow 64$, 3×3, pad 0), MaxPool, flatten (9216), FC $9216 \rightarrow 128$, FC $128 \rightarrow 10$	$1{,}199{,}882$
CNN, 9 layers (CIFAR-10)	Conv($3 \rightarrow 32$), Conv($32 \rightarrow 64$), MaxPool, Conv($64 \rightarrow 128$), Conv($128 \rightarrow 128$), MaxPool, Conv($128 \rightarrow 256$), Conv($256 \rightarrow 256$), MaxPool, flatten (4096), FC $4096 \rightarrow 1024 \rightarrow 512 \rightarrow 10$	$5{,}851{,}338$
CNN, 13 layers (CIFAR-10)	channels $32, 64$ | $128, 128$ | $256, 256, 256$ | $512, 512, 512$, MaxPool after each group; flatten (2048), FC $2048 \rightarrow 1024 \rightarrow 512 \rightarrow 10$	$10{,}244{,}042$
CNN, 15 layers (CIFAR-10)	channels $32, 64$ | $128, 128$ | $256, 256, 256, 256$ | $512, 512, 512, 512$, MaxPool after each group; flatten (2048), FC $2048 \rightarrow 1024 \rightarrow 512 \rightarrow 10$	$13{,}193{,}930$

All convolutions are 3×3. The 4-layer network uses no padding; the three CIFAR-10 networks pad by 1. Dropout in the prior network is channel-wise for the 4- and 9-layer architectures and element-wise for the 13- and 15-layer ones; both follow the reference implementation, which differs between them.

3.3 Design choices

Every design choice prioritises analysing the certificate over maximising accuracy. Four settle the study: every certificate is reported through the PAC-Bayes-kl inversion [15, 22], the tightest of the classical statements, so that the objectives differ only in the posterior they produce; the learnt prior takes the disjoint-subset route rather than the differential-privacy route of Dziugaite and Roy [5], because its independence condition is directly verifiable, at the cost of a smaller bound set; the posterior is a diagonal Gaussian, which keeps the KL a sum of per-weight closed forms; and the reference architectures are used without augmentation, pretraining or a learning-rate schedule. The departures from the reference are compute-driven rather than chosen: a per-cell grid search of its size, over five seeds, three datasets and five architectures, is beyond the compute here, so we fix one configuration and take a smaller Monte-Carlo budget, then measure that budget rather than assume it (Section 4.4). Appendix C gives the alternatives considered and rejected in each case.

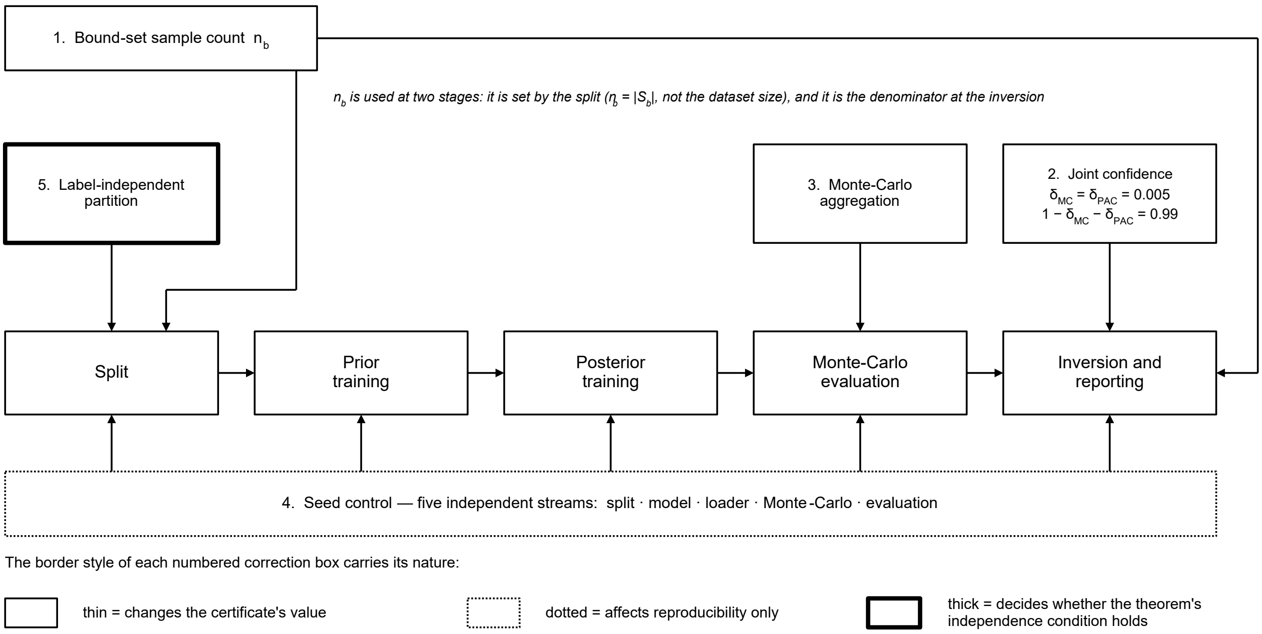


Fig. 4. Where each correction acts. The border style carries the distinction the list cannot: three change the certificate's value, one affects only reproducibility, and one decides whether the theorem's independence condition holds at all. The sample count is drawn against two stages because it is used at both, entering the split and again the inversion that reports the bound.

3.4 Corrected pipeline

All experiments use a clean reimplementation that makes the certificate bookkeeping explicit and testable. It makes five tested changes to the reference pipeline, three of which alter the reported certificate value. Figure 4 places each on the stage it acts on.

1. **Bound-set sample count.** The denominator n_{b} in (2) is $|S_{\mathrm{b}}|$, the held-out bound subset, not the size of the underlying dataset.
2. **Joint confidence.** The Monte-Carlo and PAC-Bayes steps use separate failure probabilities $\delta_{\mathrm{MC}} = \delta_{\mathrm{PAC}} = 0.005$, combined by a union bound to a reported confidence of 0.99, rather than reusing one δ across both inversions.
3. **Monte-Carlo aggregation.** The empirical Gibbs risk is accumulated example-weighted over the bound set and averaged over exactly m samples, fixing an off-by-one in the original per-batch averaging.

The fourth affects reproducibility; the fifth affects whether the learnt-prior construction satisfies the theorem's independence condition at all, though neither changes the certificate arithmetic once the split is fixed:

4. **Seed control.** Five independent seed streams (split, model, loader, Monte-Carlo, evaluation) are derived from one base seed, so the seed genuinely changes the split and the model. The evaluation stream is separate so that the descriptive test errors, which also sample posterior weights, do not inherit the RNG state the certificate's m draws left behind: otherwise raising the Monte-Carlo budget would move a test error that does not depend on it.

5. **Label-independent partition.** The subset S and its split into S_0 and S_b are drawn by a uniform random permutation rather than by taking a prefix. We first replaced the prefix split with a class-stratified one, then rejected that. Its class quotas are taken over the whole of S, which contains S_b, so S_0 comes to depend on bound-set labels; and a sample with fixed class quotas is not the i.i.d. draw the bound assumes.

All five rest on the explicit, saved bookkeeping of Section 3.8, which also keeps the six notions of error distinct: the raw Monte-Carlo 0–1 estimate, the corrected empirical Gibbs risk, the certificate, and the stochastic, posterior-mean and ensemble test errors, so the certified quantity is never silently conflated with a test error.

Each test targets a specific correction rather than basic pipeline execution, and fixes a concrete input and an expected output rather than a plausible range: the split tests assert the counts $6{,}000$ and $3{,}000$ at a tenth of the data, the aggregation test asserts agreement to 10^{-9} between a single batch and a ragged batching of a deterministic posterior, and the confidence test asserts 0.99 to 10^{-12} rather than $1 - \delta$. Appendix B lists all forty-two with their tolerances. A test that never fails defends nothing, so we checked that each bites rather than assuming it: disabling the prior-selection union in the source makes two of its five tests fail, and letting one S_0 example into the bound loader makes two of the loader tests fail. They check the bookkeeping around the bound, not the bound itself.

3.5 Training protocol

Both the prior network and the posterior are trained by SGD with momentum and no weight decay. The prior network minimises the negative log-likelihood of its labels on S_0; the posterior minimises the objective of Section 2.6. Table 2 lists every setting, and one training configuration is shared by every cell: its entries do not vary with the dataset, the architecture, the objective or the prior family. Only the evaluation batch cap differs, and for memory rather than training reasons: the bound set is scored in a single batch for the FCN cells, in batches of $2{,}500$ for the two CNN cells on 28×28 data, and $12{,}500$ on CIFAR-10, so that the deeper networks' Monte-Carlo evaluation fits in device memory. The cap changes how the estimate is batched, not what it estimates (Section 2.4).

Table 2. The training configuration shared by every cell.

Optimiser	SGD, no weight decay	Prior scale σ_0	0.03
Posterior learning rate	5×10^{-3}	Prior distribution	Gaussian
Posterior momentum	0.95	p_{min}	10^{-5}
Prior learning rate	5×10^{-3}	δ_{train}	0.025
Prior momentum	0.99	δ_{MC}, δ_{PAC}	0.005 each
Prior dropout	0.2	Monte-Carlo samples m	$2{,}000$
Batch size	250	Ensemble samples L	100
Prior epochs	70	Prior/bound split	0.5
Posterior epochs	100	KL penalty β (all objectives)	0.1

Since one fixed configuration replaces the reference's per-cell grid search, two of these values were not simply copied. The learning rates, momenta, σ_0, p_{min}, batch size and δ_{train} are those the reference reports for MNIST, and the epoch counts follow its statement that the networks converge by roughly 70 epochs. One value had to be recovered rather than copied: the reference implementation's example script ships a posterior learning rate of 10^{-3}, at which our first grid gave a stochastic test error about four times the published one, whereas the reference's text identifies 10^{-3} as converging slowly and 5×10^{-3} as the working value. We resolved that in favour of the text, but what prompted us to look was a test-set discrepancy, so the learning-rate choice was indirectly informed by the test set. It costs the certificate nothing, since it acts only on the posterior, but it does mean the reported test errors are not a wholly untouched estimate. Apart from that, the only sweep we ran ourselves was over the prior scale, $\sigma_0 \in \{0.01, 0.02, 0.05\}$ on MNIST at a reduced budget: the certificate was almost unchanged across $[0.02, 0.05]$ and degraded at 0.01, so $\sigma_0 = 0.03$ sits in a flat region rather than at a tuned optimum. Those comparisons read certificates computed on the bound set, so the prior was selected using S_{b}, and every certificate we report is corrected for that by the union of Section 2.4. The reference grid-searches the same parameter per cell without any such correction. This remains an explicitly *reduced-compute protocol*: the reference grid-searches learning rate, prior scale and dropout per cell. Section 4.4 measures the effect of the smaller m; the absent grid search is not quantified and is carried as a threat to validity (Section 5).

3.6 Seeds and statistics

Each headline cell is run over five base seeds (0–4), each genuinely changing the data split, the model initialisation, the loader order and the Monte-Carlo draws. Main-table quantities are means $\pm$ standard deviations over seeds, with per-seed values in the appendix. The across-seed variability of the training procedure is not a PAC-Bayes confidence interval.

3.7 Sensitivity and controlled-difficulty studies

Three further studies use the saved checkpoints. The **Monte-Carlo sensitivity** curve isolates the effect of the sample budget m on a fixed posterior. At each budget $m \in \{1{,}000, 2{,}000, 5{,}000, 10{,}000, 50{,}000, 150{,}000\}$, up to the reference's, we genuinely re-draw the Monte-Carlo estimate on the checkpoint and recompute the certificate. Since the raw mean is an unbiased estimate of the Gibbs risk, the only systematic route by which m enters is the finite-MC slack $\log(2/\delta_{\text{MC}})/m$ of the first KL inversion, so we also compute an analytical curve with the estimate held at its $m = 1{,}000$ value; the two agreeing is what shows the movement is slack and not a shifting estimate. The **small-data** sweep retrains the MNIST FCN f_{quad} learnt-prior cell on 100, 50, 20 and 10% of the training data, with the corrected n_{b}. The **controlled-difficulty** study applies label noise $\eta \in \{0, 0.1, 0.2, 0.4\}$ to the same cell with architecture, sample size and compute held fixed.

The noise is symmetric: each example of S is independently marked with probability η and a marked label is replaced by one drawn uniformly from the nine *other* classes, so a corrupted label is never the original and the expected fraction of wrong labels is η. It is applied once, to S as a whole and before the prior/bound split, so S_0 and S_{b} are corrupted at the same rate and the prior network also trains on noisy labels; the test set is untouched, and the noise draw has its own seed. The certificate then bounds the Gibbs risk under the noisy-label distribution, whereas the test errors in the same rows are measured against clean labels, so the two are not comparable. All three studies use the same five seeds as the headline matrix.

3.8 Compute and reproducibility

Experiments run on two NVIDIA RTX 3080 cards (20 GB each) under PyTorch 2.5.1 and CUDA 12.4 [17], and the full study comprises 115 runs. Results are aggregated from the per-run records into a single processed registry, which de-duplicates by the full configuration and records the status and any exclusion reason of every run, so that failed or non-finite runs are dropped with a stated reason rather than silently. One criterion beyond the mechanical ones is the only exclusion any run triggers. A learnt-prior cell whose prior network never leaves the chance-level plateau is not an instance of the protocol, since the posterior is then initialised at an untrained prior, so we exclude a learnt-prior run whose prior network finishes within five percentage points of chance, which for ten classes is 0.90, on S_0, the subset it was trained on. Scoring the screen there rather than on the test set keeps it a training diagnostic: it consults no held-out data and exists before any certificate does, so it cannot be tuned against a result. Separation is wide: prior networks that trained reach at most 0.228 on S_0, and the two that did not sit at 0.866 and 0.899. Every table and figure reporting experimental results in this dissertation is produced from that registry; Figures 1–4 are explanatory schematics rather than results figures, and the pre-correction outputs are retained separately as an archival audit record and are not used in any reported result.

This provenance is what allows each reported certificate to be recomputed and checked independently. Each run stores all of its seeds, the sample accounting (n_{total}, n_{selected}, n_{prior}, $n_{\text{posterior}}$, n_{b}), the three confidence parameters, the raw Monte-Carlo estimate, the KL and the resulting certificate, alongside the library versions, the hardware and the software version. Any certificate can therefore be recomputed without re-training, which is how the prior-selection union of Section 2.4 was applied to all runs after the fact. The registry carries the corrected and uncorrected values side by side, and recomputing the uncorrected column from the stored ingredients reproduces all 115 recorded certificates to 10^{-9}.

The artefact is laid out so that each of those claims can be checked from a command. One YAML file per headline cell fixes every entry of Table 2, and the published values used in Table 4 are held in a separate reference file annotated with the source column each was read from, so the comparison cannot drift onto a different column of the reference table. Appendix D gives the directory layout, the commands that regenerate a run, the matrix, the tests and every derived output, and the pinned environment.

4 Results

All tables and figures below are generated from the processed results registry, as are the values quoted in the running text.

4.1 Correctness checks

All forty-two tests of Appendix B pass, and three checks run over the results themselves. Every certificate recomputes from its saved Monte-Carlo estimate, KL and n_{b}, reproducing all 115 recorded values to 10^{-9}, and this now gates aggregation. Every saved partition recomputes from its recorded seed and matches index for index, so no reported run used a label-dependent split whatever its configuration says. And each run asserts before training that its loaders carry that partition, so the indices checked are the indices consumed. Each reported 0.99 confidence is *per certificate, per run*: a mean over seeds describes several such certificates and is not itself a 0.99 bound, which would need a further union over the rows. It is conditional in one further respect, stated in Section 2.4: the union over prior scales assumes a candidate grid fixed independently of the bound set, which we argue for but cannot demonstrate. Every 0.99 below carries that assumption.

4.2 Reproduction and main results

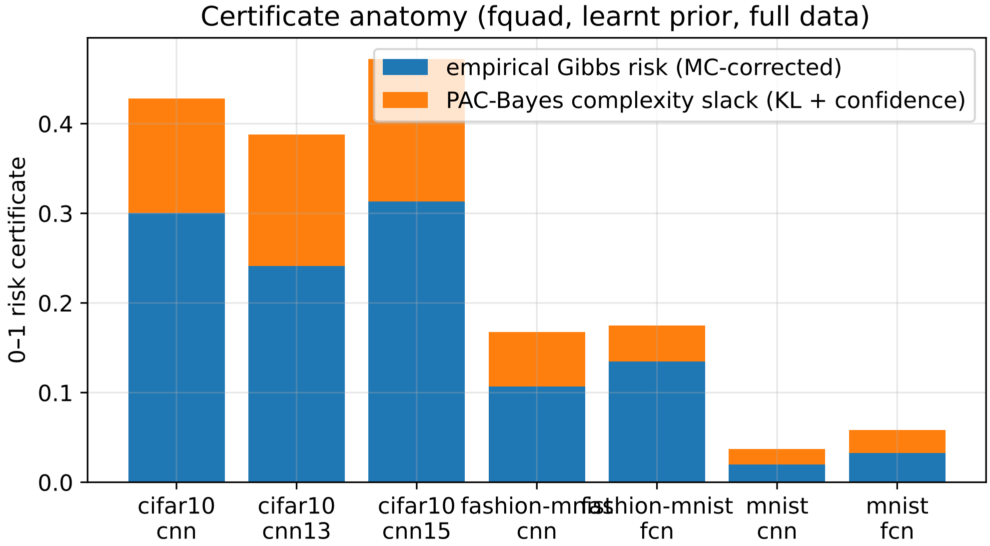


Fig. 5. Certificate anatomy: the empirical Gibbs risk and the PAC-Bayes complexity slack (KL + confidence terms) for the learnt-prior f_{quad} cells.

On MNIST with a learnt prior, the tightest 0–1 certificate is obtained by f_{classic} (0.0455), followed by f_{quad} (0.0580), f_λ (0.0777) and f_{bbb} (0.2658); all are non-vacuous and reproduce the qualitative finding of the reference that bound-minimising objectives yield far tighter certificates than the accuracy-oriented f_{bbb} baseline, whose certificate is loose precisely because it carries the largest KL term (Table 3). We do *not* observe the $f_{\text{quad}} \lesssim f_\lambda < f_{\text{classic}}$ ordering sometimes quoted: the certificate is the same PAC-Bayes-kl inversion for every objective, so the ordering reflects only which posterior each produces, and here f_{classic} yields the lowest-KL one. With the hyper-parameters fixed this is a comparison of these runs, not a general ordering of the objectives. Consistently, f_{bbb} attains the best stochastic test error on MNIST $(0.0187$ against 0.0219 for $f_{\text{quad}})$ yet the loosest certificate, because its objective is not calibrated to the reported bound.

Figure 5 shows where the looseness comes from. For the tight learnt-prior cells the certificate sits only a little above the empirical Gibbs risk: on the MNIST FCN f_{classic} cell the complexity slack KL$/n_{\text{b}}$ is around 0.0002, so the bound is almost entirely empirical risk. As the posterior is allowed a larger KL the slack grows: f_{quad} and f_λ trade slightly lower empirical risk for a noticeably larger KL, which is why they come out looser than f_{classic} at similar accuracy, and f_{bbb} is the extreme, its KL dominating the certificate even though its empirical risk is the smallest of the four. Tightness therefore depends on the empirical risk and on the KL cost of reaching it, not on accuracy alone. That also explains the cross-dataset behaviour: the empirical-risk floor rises with difficulty (Sec-

Table 3. Headline cells (mean $\pm$ SD over five seeds). Each run's certificate is a 0–1 PAC-Bayes-kl bound on the Gibbs risk holding at confidence 0.99; the column is the descriptive mean of five such per-run certificates and is not itself a 0.99 bound, which would need a further union over the five. The two test errors are descriptive deployment metrics and carry no guarantee. n_{b} is the bound-set size actually used, which halves for a learnt prior.

Dataset	Model	Prior	Obj.	n_{bound}	Emp. Gibbs	KL/n_{bound}	**Certificate**	Gibbs test	Post.mean test
CIFAR-10	CNN	learnt	f_{bbb}	25000	0.0767 ± 0.0027	3.35041	0.9801 ± 0.0024	0.2412 ± 0.0103	0.2162 ± 0.0054
CIFAR-10	CNN	learnt	f_{classic}	25000	0.3174 ± 0.0086	0.01169	0.3932 ± 0.0084	0.2902 ± 0.0091	0.2615 ± 0.0077
CIFAR-10	CNN	learnt	f_{quad}	25000	0.3003 ± 0.0084	0.03373	0.4277 ± 0.0074	0.2802 ± 0.0086	0.2525 ± 0.0077
CIFAR-10	CNN	data-free	f_{quad}	50000	0.9203 ± 0.0006	0.00029	0.9297 ± 0.0005	0.8978 ± 0.0020	0.9000 ± 0.0000
CIFAR-10	cnn13	learnt	f_{quad}	25000	0.2416 ± 0.0033	0.04760	0.3881 ± 0.0038	0.2202 ± 0.0055	0.1791 ± 0.0027
CIFAR-10	cnn15	learnt	f_{quad}	25000	0.3134 ± 0.0524	0.05188	0.4722 ± 0.0487	0.2847 ± 0.0462	0.2265 ± 0.0459
Fashion-MNIST	CNN	learnt	f_{quad}	30000	0.1069 ± 0.0014	0.01411	0.1674 ± 0.0048	0.0931 ± 0.0024	0.0866 ± 0.0011
Fashion-MNIST	FCN	learnt	f_{bbb}	30000	0.0913 ± 0.0007	0.72021	0.6632 ± 0.0045	0.1118 ± 0.0016	0.1018 ± 0.0017
Fashion-MNIST	FCN	learnt	f_{classic}	30000	0.1387 ± 0.0013	0.00087	0.1582 ± 0.0017	0.1204 ± 0.0028	0.1136 ± 0.0024
Fashion-MNIST	FCN	learnt	f_{quad}	30000	0.1349 ± 0.0011	0.00526	0.1746 ± 0.0024	0.1186 ± 0.0030	0.1109 ± 0.0035
Fashion-MNIST	FCN	data-free	f_{quad}	60000	0.2461 ± 0.0018	0.08728	0.4480 ± 0.0011	0.2254 ± 0.0062	0.1891 ± 0.0056
MNIST	CNN	learnt	f_{quad}	30000	0.0201 ± 0.0006	0.00414	0.0369 ± 0.0013	0.0108 ± 0.0011	0.0095 ± 0.0009
MNIST	FCN	learnt	f_{bbb}	30000	0.0138 ± 0.0003	0.24945	0.2658 ± 0.0065	0.0187 ± 0.0007	0.0149 ± 0.0006
MNIST	FCN	learnt	f_{classic}	30000	0.0373 ± 0.0006	0.00022	0.0455 ± 0.0008	0.0240 ± 0.0009	0.0192 ± 0.0010
MNIST	FCN	learnt	f_{λ}	30000	0.0294 ± 0.0005	0.02032	0.0777 ± 0.0013	0.0211 ± 0.0003	0.0170 ± 0.0005
MNIST	FCN	learnt	f_{quad}	30000	0.0328 ± 0.0006	0.00625	0.0580 ± 0.0011	0.0219 ± 0.0006	0.0176 ± 0.0003
MNIST	FCN	data-free	f_{quad}	60000	0.1199 ± 0.0020	0.13763	0.3496 ± 0.0028	0.0927 ± 0.0033	0.0570 ± 0.0018

tion 4.8 shows where), so even a well-controlled KL cannot pull Fashion-MNIST or CIFAR-10 into the MNIST range.

Comparison with the published numbers. Table 4 sets our values against the reference's for the four MNIST cells named as reproduction targets in Section 1.4, using the same-definition columns recorded in the reference file of Section 3.8. All four reproduce on the test-error metric: the stochastic test error differs from the published value by between -0.0024 and $+0.0033$, inside the one-percentage-point tolerance, and the sign of the difference varies between cells rather than pointing one way, as it would if our posteriors were systematically better or worse trained. That is evidence about predictive behaviour, not about each component: the test error is end-to-end, and the component claims rest on the tests of Appendix B.

Table 4. Reproduction of the four MNIST cells named in Section 1.4, against the published values of Pérez-Ortiz et al. [20]. Ours are means over five seeds. The published numbers were obtained with $m = 150{,}000$ Monte-Carlo samples and a per-cell grid search; ours with $m = 2{,}000$ and one fixed configuration.

			Certificate			Stochastic test error		
Model	Prior	Obj.	Published	Ours	Δ	Published	Ours	Δ
FCN	data-free	f_{quad}	0.3155	0.3496	$+0.0341$	0.0951	0.0927	-0.0024
FCN	learnt	f_{quad}	0.0279	0.0580	$+0.0301$	0.0204	0.0219	$+0.0015$
FCN	learnt	f_{λ}	0.0354	0.0777	$+0.0423$	0.0178	0.0211	$+0.0033$
CNN	learnt	f_{quad}	0.0155	0.0369	$+0.0214$	0.0127	0.0108	-0.0019

The certificates do not meet the same tolerance: ours are looser in all four cells, by between $+0.0214$ and $+0.0423$. The Monte-Carlo sweep of Section 4.4 measures how much of that is budget: re-drawing at the reference's $m = 150{,}000$ takes the MNIST FCN f_{quad} seed-0 checkpoint from 0.0572 to 0.0420, removing a little over half of that checkpoint's 0.0293 gap. The remainder is unexplained. The reproduction therefore succeeds on the predictive metrics and is partial on the certificate, with one protocol difference measured and the rest identified but not bounded.

The standard deviations in Table 3 are small relative to the gaps the conclusions rest on: the three bound-minimising MNIST learnt-prior certificates vary by one to two thousandths across the five seeds, whereas the gap between objectives, or between a learnt and a random prior, is far larger. That the spread is measurable at all is the seed correction: in the original pipeline the split and the initialisation were reset to fixed constants regardless of the command-line seed, so every "seed" produced an identical run. With five streams genuinely driving the split, the initialisation, the loader order, the Monte-Carlo draws and the evaluation, the spread we report is real, and small enough that the orderings we draw conclusions from are not artefacts of one lucky run. It stays separate from the PAC-Bayes confidence, which is about one trained model rather than variation between runs.

4.3 Prior comparison

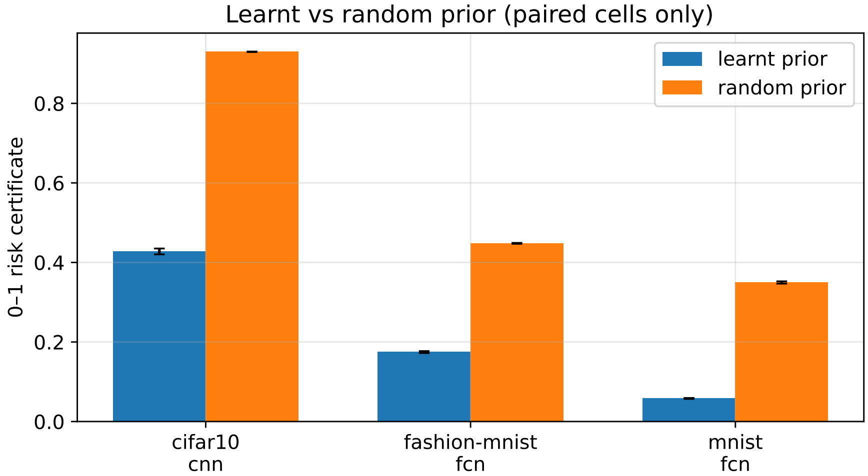


Fig. 6. Learnt vs random prior certificate, shown only for cells where both priors were run (including the CIFAR-10 control).

The prior changes the KL a useful posterior must pay, and moves certificate tightness a long way in every paired control (Figure 6). On the MNIST FCN f_{quad} cell, going from a random to a learnt prior tightens the certificate from 0.3496 to 0.0580 while KL$(Q\|P)$ falls from 8258.0648 to 187.4999 nats and the empirical Gibbs risk from 0.1199 to 0.0328, so accuracy improves alongside the certificate. The same direction holds on Fashion-MNIST ($0.4480 \to 0.1746$).

Which of the two terms does the work is a separate question, and the paired cells answer it differently. The quantity to compare is not the raw KL but KL$/n_{\text{b}}$, since n_{b} halves when the prior is learnt. On MNIST the complexity term falls by 0.131 against 0.087 for the empirical risk, so the complexity route is the larger. On Fashion-MNIST the two are comparable and the ranking reverses: 0.082 against 0.111. On CIFAR-10 the complexity term *rises* by 0.033, because the random-prior posterior barely leaves its initialisation and so pays almost no KL, and the whole of the gain is the empirical risk falling from 0.9203 to 0.3003. The learnt prior tightens the certificate in all three, then, but not by one mechanism: across these three paired cells the dominant contribution shifts from the complexity term to the empirical risk. We do not attribute that shift to difficulty, since architecture and budget change along the same sequence. The comparison is between two whole protocols rather than a pure prior-family intervention: the random prior is independent of all of S and so uses $n_{\text{b}} = |S|$, while the learnt prior uses the smaller $n_{\text{b}} = |S \setminus S_0|$ — and still attains the tighter certificate *despite* that handicap. CIFAR-10 is the clearest case: a random prior leaves the f_{quad} certificate at 0.9297, whereas the learnt prior moves it to 0.4277, formally non-vacuous but still loose.

Architecture. Where a CNN and an FCN are compared on the same dataset, the convolutional model gives the tighter certificate: on MNIST the f_{quad} learnt-prior certificate is 0.0369 (CNN) versus

0.0580 (FCN), and on Fashion-MNIST 0.1674 versus 0.1746.

Depth on CIFAR-10. The reference obtains its tighter CIFAR-10 certificates with deeper networks than the nine-layer one used above, so we ran its 13- and 15-layer architectures at the same five seeds and the same fixed configuration. Within this f_{quad} depth sweep the mean certificate improves from nine to thirteen layers and worsens at fifteen: 0.4277, 0.3881 and 0.4722 respectively. The 13-layer network also improves the deployed error (0.1791 versus 0.2525), and the gain concentrates where the error is: the per-class Gibbs risk of Section 4.8 falls from 0.448 to 0.340 on *cat*, the hardest class. It is also the tightest CIFAR-10 certificate in the study, narrowly ahead of f_{classic} at nine layers (0.3932), and so the best of the three depths tested under f_{quad} and this configuration.

The 15-layer network does not continue the trend, for an instructive reason. Two of its five seeds barely left the chance-level plateau, their prior networks finishing at 0.866 and 0.899 error on the S_0 they were fitted to, against a chance level of 0.900: the second is chance to three decimals and the first sits inside the criterion's five-point tolerance, both far from the at most 0.228 every other learnt prior reaches. The posterior is then initialised at an essentially untrained prior, so the cell is not an instance of the protocol. Both are registered as `not_converged` and excluded by the criterion of Section 3.8; the remaining three give 0.4722, looser than the 13-layer result and far more variable. That mean is conditional on convergence, so the failure rate is part of the result: at fifteen layers this protocol trains 3 of 5 seeds, against 5 of 5 everywhere else in the study. Twelve convolutional layers with no normalisation, no warm-up and one fixed learning rate is past what it trains reliably — a limit of the fixed reduced-compute configuration rather than of the method, and the price of holding one configuration across every cell.

Across datasets. The learnt-prior f_{quad} certificate rises from MNIST (0.0580, FCN) through Fashion-MNIST (0.1746, FCN) to CIFAR-10 (0.4277, 9-layer CNN). The pattern is descriptive, since architecture and compute budget change along it; Section 4.6 is the controlled test of difficulty.

4.4 Monte-Carlo sensitivity

The comparison above attributed our looser certificates partly to the Monte-Carlo budget; this sweep measures that. Holding the posterior fixed at a saved checkpoint and re-drawing at each budget, the certificate tightens monotonically as m grows and flattens well before the reference's budget (Figure 7). On the MNIST FCN f_{quad} checkpoint it falls from 0.0654 at $m = 1{,}000$ to 0.0572 at our $m = 2{,}000$ and 0.0420 at the reference's $m = 150{,}000$ — all the *same* seed-0 checkpoint, which is why the $m = 2{,}000$ entry is not the five-seed mean 0.0580 of Table 3. Against the published

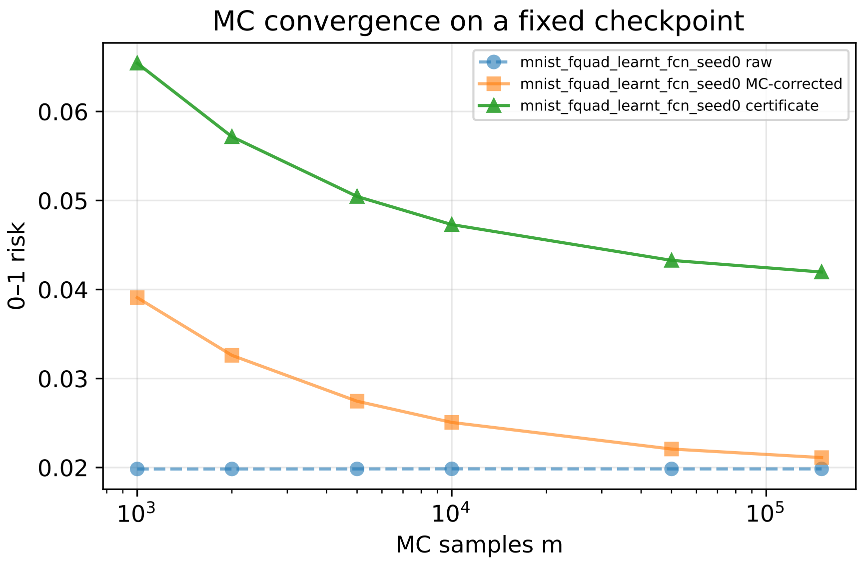


Fig. 7. Final certificate versus the Monte-Carlo budget m on a fixed checkpoint with the Monte-Carlo estimate genuinely re-drawn at each budget, so both the estimate and the finite-MC slack are free to move.

0.0279, the re-draw recovers 0.0152 of this checkpoint's 0.0293 gap. The rest is unexplained; the unmatched grid search and epoch budget are plausible contributors we did not test separately.

Two checks confirm that what moves is the slack and not the estimate. The raw Monte-Carlo estimate is stable across the whole range (0.01983 at $m = 1{,}000$ against 0.01984 at $m = 150{,}000$), as an unbiased estimator should be; and the same curve computed analytically, holding that estimate fixed, agrees with the measured one to within 2.2×10^{-5} at every budget. Both curves are in the artefact. A re-draw could have moved the certificate either way; here it fell monotonically at every step, so for this checkpoint and these draws the smaller m was the conservative one.

4.5 Small-data behaviour

Table 5. MNIST FCN f_{quad}/learnt versus training-data fraction, with the corrected bound-set size n_{b} (mean $\pm$ SD over seeds).

% data	$n_{\text{posterior}}$	n_{bound}	Emp. Gibbs	**Certificate**
100%	60000	30000	0.0328 ± 0.0006	0.0580 ± 0.0011
50%	30000	15000	0.0420 ± 0.0024	0.0777 ± 0.0049
20%	12000	6000	0.0624 ± 0.0034	0.1196 ± 0.0040
10%	6000	3000	0.0771 ± 0.0035	0.1520 ± 0.0065

The certificate loosens monotonically as the training set shrinks (Table 5 and Figure 8): from 0.0580 at full data to 0.0777 (50%), 0.1196 (20%) and 0.1520 at 10% of the data ($n_{\text{b}} = 3{,}000$). The certificate increases monotonically as the data fraction falls, and each step exceeds the across-seed spread at that setting. The small-data runs show most directly why the sample-count audit matters. The certificate denominator is the bound-set size n_{b}, which at 10% data is $3{,}000$, whereas taking it to be

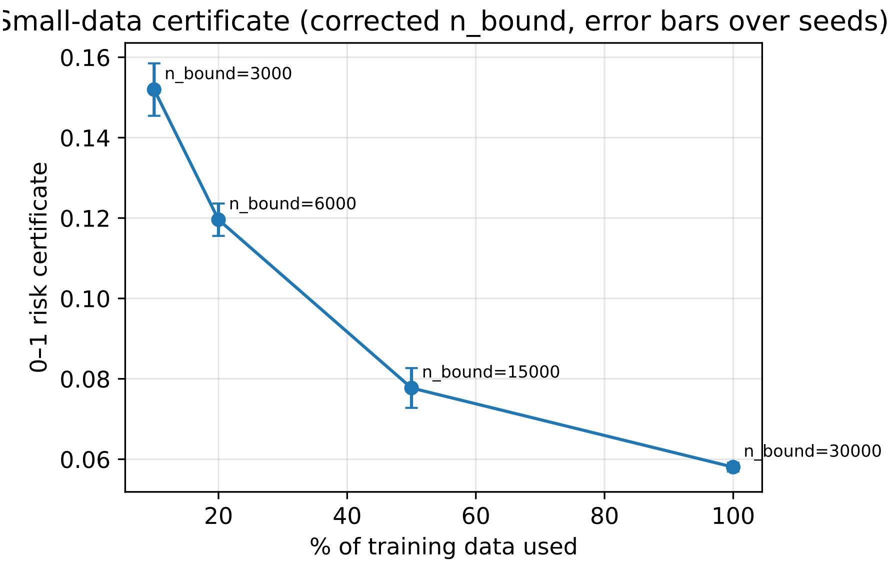


Fig. 8. Certificate versus training-data fraction with the corrected n_b and across-seed error bars.

the size of the full training set would instead give $60{,}000$. Using the full dataset size understates the complexity term and reports a spuriously tight certificate at small n: on these same five checkpoints the wrong denominator returns 0.092 where the correct one returns 0.1520, hiding most of the degradation. The bound stays non-vacuous down to 10% of the data, so the degradation is graceful, but it is far from the near-flat behaviour a mis-counted n would suggest.

4.6 Controlled difficulty

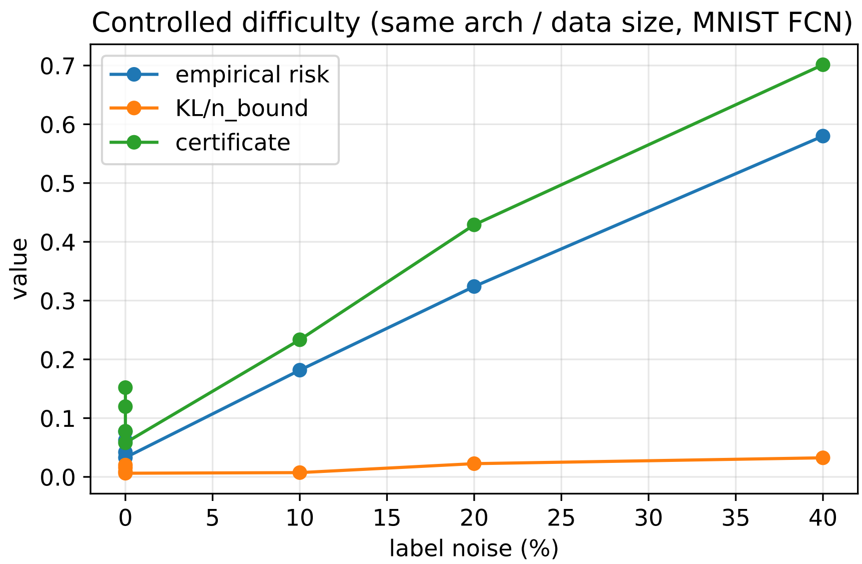


Fig. 9. Empirical risk, KL$/n_\mathrm{b}$ and certificate versus label-noise level for the MNIST FCN f_quad/learnt cell (architecture, sample size and compute held fixed).

The cross-dataset trend above could not separate difficulty from the architecture and budget that change alongside it; this study removes that confound. With architecture, sample size and compute held fixed and label noise injected into the MNIST FCN f_{quad} learnt-prior cell, the certificate rises monotonically with the noise level (Figure 9), from 0.0580 at no noise to 0.2334, 0.4289 and 0.7013 at 10, 20 and 40% noise. Successive noise levels are separated by far more than the across-seed spread (Figure 9). The increase is driven mainly by the empirical-risk term, since corrupted labels raise the error floor that any posterior must pay on S_{b}, with a smaller rise in KL$/n_{\text{b}}$. Because every other factor is held fixed, this is the one place we attribute a difficulty effect to the intervention itself. The certified quantity here is the Gibbs risk on the *noisy-label* distribution the model is trained and evaluated against, not on clean labels.

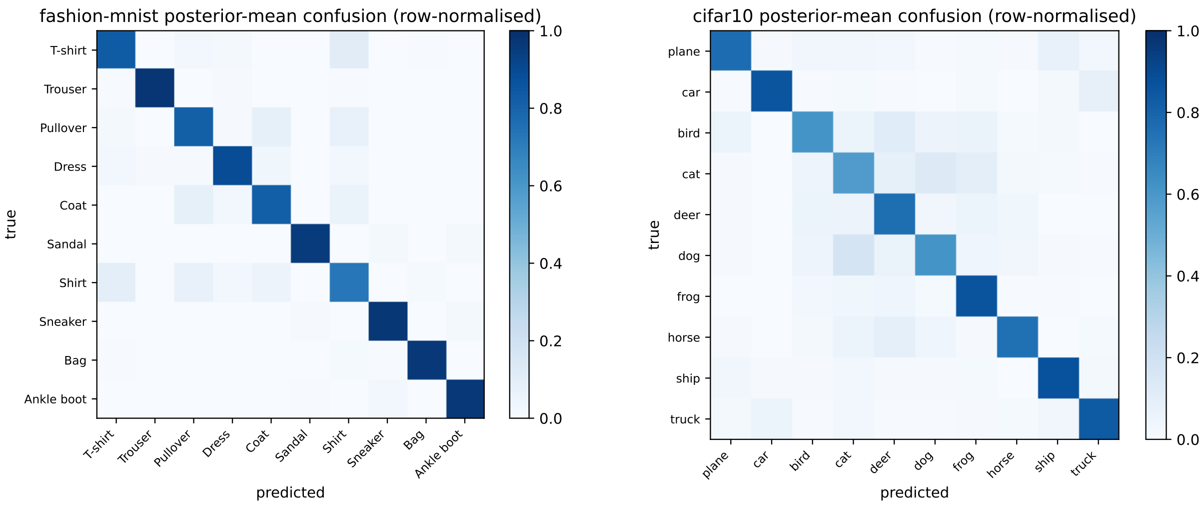
4.7 The certified predictor versus the deployed one

Table 3 also lets us compare the predictor the certificate covers against the one a user would deploy. The posterior-mean test error is below the stochastic (Gibbs) error in every learnt-prior cell, though not universally: on the CIFAR-10 data-free-prior cell, where neither predictor has learnt much, the ordering reverses. The finite ensemble is usually below the stochastic error but is not uniformly below the posterior mean. On the MNIST FCN f_{quad} cell the stochastic error is 0.0219 against a posterior mean of 0.0176, and the same direction holds on Fashion-MNIST and CIFAR-10 with a learnt prior. The gap is consistent across the learnt-prior cells and not negligible, at roughly a fifth of the stochastic error on the MNIST cell, and by Section 2.3 it does not transfer the certificate to the posterior mean, so a user needing a guarantee for the deterministic network does not have one here. Closing that gap is the clearest theoretical extension of the project.

4.8 Where the error falls

The certificate bounds an aggregate risk, and it is informative to see how the error distributes across classes, since that is what the empirical-risk term integrates over. Figure 10 shows the posterior-mean confusion matrices on the two harder datasets. On Fashion-MNIST the error concentrates in the upper-body garments, the worst class being *shirt*, confused mainly with *pullover*, *coat* and *T-shirt*, while the footwear classes are almost perfectly separated; on CIFAR-10 it sits in *cat*, *dog* and *bird*, with the rigid vehicle classes much easier. Those matrices describe the posterior-mean predictor on the test set, which is not the object the certificate bounds, so we also decomposed the certified quantity itself: the Monte-Carlo Gibbs risk on S_{b}, accumulated by true class over the same $m = 2{,}000$ draws that produce the certificate, on the seed-0 checkpoint of each cell. What is decomposed is the Monte-Carlo average $\bar{R}$, before the finite-sample correction of (4), so the per-class values below sit on that scale rather than on the scale of the corrected $\overline{R}^{+}$ quoted elsewhere. Its class average reproduces the run's own aggregate estimate to within 10^{-4}, which is the check that the decomposition is of the right quantity.

The certified risk concentrates in the same places, and more sharply: on Fashion-MNIST the Gibbs risk on *shirt* is 0.293 against a dataset-wide 0.108 on the same scale, and on CIFAR-10 it is *cat* 0.448, *dog* 0.383 and *bird* 0.366 against 0.264 for the nine-layer network. The worst class carries close to three times the average risk the bound integrates over on Fashion-MNIST and about 1.7 times it on CIFAR-10. That is what the empirical-risk term of Section 4.2 is made of, and why a well-controlled KL cannot pull the harder datasets into the MNIST range. It also marks a boundary of the method: the certificate is one number over all classes, so a deployment that cares about the worst class is not served by it.



(a) Fashion-MNIST

(b) CIFAR-10

Fig. 10. Posterior-mean confusion matrices (row-normalised) on the two harder datasets. The error concentrates in visually similar classes, which raises the empirical Gibbs risk that the certificate is built on.

5 Evaluation and Reflection

5.1 Evaluation strategy

The evaluation is organised around the questions of the introduction rather than a single accuracy number, because the object of study is the certificate and its trustworthiness. Correctness is checked before any result is interpreted: each audited condition has a unit test, and every certificate is recomputed from its saved ingredients. Every cell runs five seeds and is reported as a mean with a standard deviation, the sole exception being the CIFAR-10 15-layer cell, where two runs did not train. The two claims easiest to confound, the effect of the prior and of difficulty, rest on controlled comparisons instead: a paired learnt-versus-random prior on every dataset, and a label-noise sweep with architecture and sample size held fixed. Causal claims are restricted to those.

5.2 Threats to validity

The main threats concern differences from the reference protocol and limits on how the results can be interpreted. On the first, the reduced-compute protocol of Section 3.5 deviates in several ways and only one of them, the Monte-Carlo budget, has been measured (Section 4.4); the absent hyper-parameter search and the epoch budget can move the certificate either way and we do not bound them. One threat is internal to our own correction: the union over prior scales needs a candidate set fixed independently of the bound set, and ours was written down afterwards (Section 2.4). The exposure is small, since the grid covers every scale the probe could plausibly have returned, but that is an assumption about our procedure, not something the data confirms.

The interpretive limits concern the certified predictor, the cross-dataset comparison, the prior comparison and seed variability. Only the Gibbs classifier is certified (Section 2.3); a formal bound for the deterministic predictors via the C-bound is left to future work. As noted in Section 4.2, the cross-dataset comparison is descriptive because its confounders are not controlled, and the label-noise study is where a difficulty claim is made instead. Even the prior comparison, our most controlled effect, compares two whole protocols rather than one isolated factor, which is why a CIFAR-10 random-prior control is included rather than inferred from the easier datasets. Finally, the across-seed spread is the variability of the training procedure and must not be read as a confidence interval on the certificate: the PAC-Bayes guarantee is about one trained model, holding with probability at least $1 - \delta_{\text{MC}} - \delta_{\text{PAC}} = 0.99$.

5.3 Limitations

A non-vacuous certificate is not necessarily a useful one: a CIFAR-10 certificate below 1 may still be too loose to drive a deployment decision. Depth is varied only in the CIFAR-10 f_{quad} sweep and never with width; we test no modality beyond image benchmarks, and implement neither the differentially-private-prior nor the majority-vote route. The audit corrects and tests the pipeline issues we identified, but is not an exhaustive verification of the upstream code.

5.4 Reflection on the project process

The most useful lesson from the project was how easily a reproduction of a certificate can be wrong in a way that nothing flags. The original difficulty ladder appeared to hold, the small-data certificate appeared to stay tight, and the objective ordering appeared to match the literature, until each was traced back to a specific step in the pipeline. The sample-count convention is the clearest example: at 10% data, putting the full $60{,}000$ in the denominator instead of the true $n_{\mathrm{b}} = 3{,}000$ makes a certificate that is really 0.152 read as 0.092, and either figure looks plausible. That experience shifted the project towards verifiable certificates, and motivated building the unit tests and the registry before the headline runs.

With hindsight, one thing would have been done much earlier: profiling. The data pipeline, not the network, dominated wall-clock time, and the Monte-Carlo evaluation ran in far smaller batches than the hardware needed. Fixing both left the reported numbers untouched, since neither changes what is computed, but it turned the full matrix from a budget question into a routine one. The registry, the per-run provenance and the regeneration of every results figure from raw results are what make these certificates independently checkable, and on this evidence they belong in the result rather than after it.

6 Conclusion

This dissertation partially reproduced selected PAC-Bayes-with-Backprop cells on MNIST under a reduced-compute protocol, extended them to Fashion-MNIST and CIFAR-10, and re-implemented the certificate-computation pipeline with corrected bookkeeping.

On reproduction, the corrected pipeline matches the reference's MNIST stochastic test errors to within 0.0033 on all four target cells, with the sign of the difference varying between them. The certificates are uniformly looser; the Monte-Carlo sweep accounts for roughly half of the gap on the cell examined in detail, and the rest coincides with protocol differences, chiefly the absent per-cell grid search, that we identified but did not measure. The reproduction is therefore complete on the predictive metrics and partial on the certificate. Decomposing it revealed effects that test error alone does not: it separates into the empirical Gibbs risk and a complexity term scaling with KL$/n_{\mathrm{b}}$, and a learnt prior tightens it in every paired comparison we ran, though not by the same route each time. Measured on the scale that enters the bound, KL$/n_{\mathrm{b}}$, the complexity term is the larger route only on MNIST; on Fashion-MNIST the two routes are comparable, and on CIFAR-10 the learnt prior pays *more* complexity and wins entirely on empirical risk. That decomposition is what makes the certificate sensitive in the ways the sensitivity analysis probes — to the prior, the bound-set size, the Monte-Carlo budget and the seed — and it is where the audit proves its worth, since correcting the bound-set sample count alone loosens the small-data certificates substantially. Difficulty was therefore tested in the one design that controls for it: with everything but the label noise held constant, the noise raises the empirical-risk term and, through the PAC-Bayes-kl inversion, the certificate, so the effect can be attributed to the perturbation itself, with the proviso that the certified quantity is then the Gibbs risk on the noisy-label distribution.

Together these results show that PAC-Bayes-with-Backprop delivered formally non-vacuous certificates in all 23 reported cells, none reaching 1 — a statement about runs that trained, since the 15-layer cell is the mean of its three converged seeds. Non-vacuous and useful are different thresholds, though: the loosest, CIFAR-10 under f_{bbb}, is 0.9801, certifying almost nothing a practitioner would act on, and tight certificates arrived only on the easier benchmarks. In all three paired f_{quad} controls the data-dependent prior produced a large tightening — not the largest effect in the study, since 40% label noise moves the same MNIST certificate about twice as far. The value of any such number depends on the implementation that produced it, and several conventions in the reference required correction in our re-implementation. The main contribution of the project is that corrected, tested pipeline: reliable certificates require correct sample counts, confidence accounting, Monte-Carlo aggregation and seed handling, together with enough provenance to re-derive each reported value.

The clearest direction for further work is to close the gap between the certified Gibbs classifier and the deterministic predictor actually deployed. The C-bound of Lacasse et al. [11] and Germain et al. [8] relates the majority-vote risk to the mean and variance of the Gibbs risk, and adapting it to the objectives used here would give a formal guarantee for a deterministic predictor. Which one matters: the C-bound governs the majority vote induced by Q, the limit of the ensemble predictor, not the posterior-mean network whose error we report, so a guarantee for the latter would remain open. On the empirical side, the full-budget re-draw reported here removes one identified source of the discrepancy but not the absent grid search, which is what a reference-matched hyper-parameter search would settle. The CIFAR-10 results invite two extensions: the deeper 13- and 15-layer architectures that the reference shows give tighter bounds, and the differentially-private prior route, which would let the prior use all of the data and is the more promising way to attack the high empirical risk that, as the error analysis shows, is what keeps CIFAR-10 certificates loose. Finally, the controlled-difficulty design could be broadened from label noise to input corruption and class overlap, to test whether the empirical-risk mechanism identified here is the dominant one across different notions of difficulty.

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Appendices

A Per-seed complete results

Table 6 gives per-seed results for all 113 reported runs, generated from the results registry. Each row is one run, whose configuration, data split, checkpoints and metrics are retained and whose inclusion or exclusion is recorded in the registry. The registry holds 115 runs; the two that do not appear here are the learnt-prior runs whose prior network never left the chance-level plateau, at 0.866 and 0.899 error on S_0, excluded under the criterion stated in Section 3.8.

The **Condition** column identifies which study a row belongs to. *Headline* rows use the full training set with clean labels and are the cells of Table 3, run at five seeds (0–4). *Data* p% rows are the small-data sweep of Section 4.5, retrained on a uniformly random p% of the training set drawn without reading any label, at five seeds (0–4). *Noise* η% rows are the controlled-difficulty study of Section 4.6, with a fraction η of the labels of S corrupted to a uniformly random different class, also at five seeds. Rows sharing a dataset, model, prior, objective and condition differ only in their seed.

Column n_b is the bound-set size actually used in the certificate, so it varies with both the prior family and the training-data fraction. The risk certificate is the 0–1 PAC-Bayes-kl bound on the Gibbs risk at confidence 0.99, per certificate and per run. "Stoch / PM / Ens" are the stochastic, posterior-mean and ensemble test errors, which are descriptive deployment metrics and carry no guarantee; for the noise rows they are measured against clean test labels, whereas the certificate in the same row refers to the corrupted-label distribution, so the two are not comparable within those rows.

Table 6. Per-seed results for all 113 reported runs, one row per run.

Dataset	Model	Prior	Obj.	Condition	Seed	n_b	Cert	Stoch	PM	Ens
CIFAR-10	CNN	learnt	f_bbb	headline	0	25000	0.9796	0.2445	0.2181	0.2055
CIFAR-10	CNN	learnt	f_bbb	headline	1	25000	0.9817	0.2400	0.2163	0.2030
CIFAR-10	CNN	learnt	f_bbb	headline	2	25000	0.9804	0.2382	0.2181	0.2029
CIFAR-10	CNN	learnt	f_bbb	headline	3	25000	0.9824	0.2557	0.2214	0.2094
CIFAR-10	CNN	learnt	f_bbb	headline	4	25000	0.9762	0.2274	0.2072	0.1947
CIFAR-10	CNN	learnt	f_classic	headline	0	25000	0.3894	0.2865	0.2610	0.2391
CIFAR-10	CNN	learnt	f_classic	headline	1	25000	0.3996	0.2940	0.2692	0.2510
CIFAR-10	CNN	learnt	f_classic	headline	2	25000	0.3950	0.2918	0.2558	0.2408
CIFAR-10	CNN	learnt	f_classic	headline	3	25000	0.4015	0.3014	0.2691	0.2528
CIFAR-10	CNN	learnt	f_classic	headline	4	25000	0.3808	0.2771	0.2523	0.2355
CIFAR-10	CNN	learnt	f_quad	headline	0	25000	0.4244	0.2788	0.2564	0.2314
CIFAR-10	CNN	learnt	f_quad	headline	1	25000	0.4336	0.2819	0.2581	0.2403
CIFAR-10	CNN	learnt	f_quad	headline	2	25000	0.4282	0.2826	0.2460	0.2306
CIFAR-10	CNN	learnt	f_quad	headline	3	25000	0.4351	0.2907	0.2596	0.2430

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Dataset	Model	Prior	Obj.	Condition	Seed	n_{b}	Cert	Stoch	PM	Ens
CIFAR-10	CNN	learnt	f_{quad}	headline	4	25000	0.4170	0.2671	0.2425	0.2258
CIFAR-10	CNN	data-free	f_{quad}	headline	0	50000	0.9291	0.8957	0.9000	0.8942
CIFAR-10	CNN	data-free	f_{quad}	headline	1	50000	0.9302	0.9006	0.9000	0.8966
CIFAR-10	CNN	data-free	f_{quad}	headline	2	50000	0.9295	0.8960	0.9000	0.8994
CIFAR-10	CNN	data-free	f_{quad}	headline	3	50000	0.9294	0.8989	0.9000	0.9033
CIFAR-10	CNN	data-free	f_{quad}	headline	4	50000	0.9304	0.8977	0.9000	0.8984
CIFAR-10	cnn13	learnt	f_{quad}	headline	0	25000	0.3849	0.2174	0.1794	0.1717
CIFAR-10	cnn13	learnt	f_{quad}	headline	1	25000	0.3934	0.2247	0.1817	0.1718
CIFAR-10	cnn13	learnt	f_{quad}	headline	2	25000	0.3849	0.2160	0.1757	0.1700
CIFAR-10	cnn13	learnt	f_{quad}	headline	3	25000	0.3864	0.2154	0.1770	0.1682
CIFAR-10	cnn13	learnt	f_{quad}	headline	4	25000	0.3908	0.2273	0.1818	0.1747
CIFAR-10	cnn15	learnt	f_{quad}	headline	1	25000	0.5284	0.3380	0.2795	0.2753
CIFAR-10	cnn15	learnt	f_{quad}	headline	2	25000	0.4425	0.2604	0.2025	0.2037
CIFAR-10	cnn15	learnt	f_{quad}	headline	3	25000	0.4456	0.2556	0.1976	0.1955
Fashion-MNIST	CNN	learnt	f_{quad}	headline	0	30000	0.1658	0.0924	0.0864	0.0847
Fashion-MNIST	CNN	learnt	f_{quad}	headline	1	30000	0.1739	0.0932	0.0870	0.0838
Fashion-MNIST	CNN	learnt	f_{quad}	headline	2	30000	0.1627	0.0908	0.0850	0.0805
Fashion-MNIST	CNN	learnt	f_{quad}	headline	3	30000	0.1638	0.0971	0.0881	0.0837
Fashion-MNIST	CNN	learnt	f_{quad}	headline	4	30000	0.1708	0.0920	0.0867	0.0832
Fashion-MNIST	FCN	learnt	f_{bbb}	headline	0	30000	0.6616	0.1111	0.1025	0.1001
Fashion-MNIST	FCN	learnt	f_{bbb}	headline	1	30000	0.6656	0.1097	0.1000	0.0996
Fashion-MNIST	FCN	learnt	f_{bbb}	headline	2	30000	0.6573	0.1125	0.1018	0.0993
Fashion-MNIST	FCN	learnt	f_{bbb}	headline	3	30000	0.6692	0.1139	0.1042	0.0997
Fashion-MNIST	FCN	learnt	f_{bbb}	headline	4	30000	0.6622	0.1119	0.1005	0.0978
Fashion-MNIST	FCN	learnt	f_{classic}	headline	0	30000	0.1574	0.1228	0.1148	0.1122
Fashion-MNIST	FCN	learnt	f_{classic}	headline	1	30000	0.1593	0.1178	0.1125	0.1084
Fashion-MNIST	FCN	learnt	f_{classic}	headline	2	30000	0.1556	0.1224	0.1166	0.1139
Fashion-MNIST	FCN	learnt	f_{classic}	headline	3	30000	0.1586	0.1219	0.1137	0.1110
Fashion-MNIST	FCN	learnt	f_{classic}	headline	4	30000	0.1600	0.1170	0.1103	0.1065
Fashion-MNIST	FCN	learnt	f_{quad}	headline	0	30000	0.1740	0.1208	0.1130	0.1110
Fashion-MNIST	FCN	learnt	f_{quad}	headline	1	30000	0.1763	0.1162	0.1089	0.1061
Fashion-MNIST	FCN	learnt	f_{quad}	headline	2	30000	0.1713	0.1207	0.1143	0.1092
Fashion-MNIST	FCN	learnt	f_{quad}	headline	3	30000	0.1741	0.1209	0.1124	0.1088
Fashion-MNIST	FCN	learnt	f_{quad}	headline	4	30000	0.1776	0.1145	0.1057	0.1040
Fashion-MNIST	FCN	data-free	f_{quad}	headline	0	60000	0.4487	0.2219	0.1979	0.1828
Fashion-MNIST	FCN	data-free	f_{quad}	headline	1	60000	0.4463	0.2184	0.1890	0.1800
Fashion-MNIST	FCN	data-free	f_{quad}	headline	2	60000	0.4489	0.2231	0.1898	0.1803
Fashion-MNIST	FCN	data-free	f_{quad}	headline	3	60000	0.4478	0.2334	0.1851	0.1824
Fashion-MNIST	FCN	data-free	f_{quad}	headline	4	60000	0.4485	0.2303	0.1836	0.1813
MNIST	CNN	learnt	f_{quad}	headline	0	30000	0.0373	0.0095	0.0082	0.0083
MNIST	CNN	learnt	f_{quad}	headline	1	30000	0.0369	0.0107	0.0103	0.0100
MNIST	CNN	learnt	f_{quad}	headline	2	30000	0.0364	0.0124	0.0103	0.0105
MNIST	CNN	learnt	f_{quad}	headline	3	30000	0.0389	0.0110	0.0096	0.0102
MNIST	CNN	learnt	f_{quad}	headline	4	30000	0.0352	0.0104	0.0092	0.0087
MNIST	FCN	learnt	f_{bbb}	headline	0	30000	0.2657	0.0191	0.0151	0.0154
MNIST	FCN	learnt	f_{bbb}	headline	1	30000	0.2631	0.0175	0.0149	0.0148

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Dataset	Model	Prior	Obj.	Condition	Seed	n_b	Cert	Stoch	PM	Ens
MNIST	FCN	learnt	f_bbb	headline	2	30000	0.2585	0.0186	0.0141	0.0140
MNIST	FCN	learnt	f_bbb	headline	3	30000	0.2762	0.0191	0.0147	0.0145
MNIST	FCN	learnt	f_bbb	headline	4	30000	0.2653	0.0192	0.0156	0.0155
MNIST	FCN	learnt	f_classic	headline	0	30000	0.0450	0.0247	0.0206	0.0204
MNIST	FCN	learnt	f_classic	headline	1	30000	0.0448	0.0237	0.0185	0.0193
MNIST	FCN	learnt	f_classic	headline	2	30000	0.0450	0.0230	0.0197	0.0194
MNIST	FCN	learnt	f_classic	headline	3	30000	0.0465	0.0235	0.0191	0.0191
MNIST	FCN	learnt	f_classic	headline	4	30000	0.0461	0.0253	0.0182	0.0189
MNIST	FCN	learnt	f_λ	headline	0	30000	0.0766	0.0214	0.0162	0.0169
MNIST	FCN	learnt	f_λ	headline	1	30000	0.0767	0.0209	0.0173	0.0172
MNIST	FCN	learnt	f_λ	headline	2	30000	0.0768	0.0206	0.0171	0.0163
MNIST	FCN	learnt	f_λ	headline	3	30000	0.0795	0.0211	0.0167	0.0171
MNIST	FCN	learnt	f_λ	headline	4	30000	0.0785	0.0214	0.0176	0.0163
MNIST	FCN	learnt	f_quad	data 10%	0	3000	0.1487	0.0623	0.0544	0.0537
MNIST	FCN	learnt	f_quad	data 10%	1	3000	0.1628	0.0630	0.0537	0.0544
MNIST	FCN	learnt	f_quad	data 10%	2	3000	0.1488	0.0628	0.0562	0.0570
MNIST	FCN	learnt	f_quad	data 10%	3	3000	0.1532	0.0631	0.0541	0.0566
MNIST	FCN	learnt	f_quad	data 10%	4	3000	0.1463	0.0648	0.0571	0.0583
MNIST	FCN	learnt	f_quad	data 20%	0	6000	0.1268	0.0452	0.0365	0.0397
MNIST	FCN	learnt	f_quad	data 20%	1	6000	0.1178	0.0482	0.0383	0.0392
MNIST	FCN	learnt	f_quad	data 20%	2	6000	0.1183	0.0480	0.0407	0.0414
MNIST	FCN	learnt	f_quad	data 20%	3	6000	0.1172	0.0466	0.0391	0.0397
MNIST	FCN	learnt	f_quad	data 20%	4	6000	0.1178	0.0459	0.0395	0.0401
MNIST	FCN	learnt	f_quad	data 50%	0	15000	0.0699	0.0297	0.0239	0.0250
MNIST	FCN	learnt	f_quad	data 50%	1	15000	0.0830	0.0315	0.0244	0.0252
MNIST	FCN	learnt	f_quad	data 50%	2	15000	0.0777	0.0280	0.0218	0.0233
MNIST	FCN	learnt	f_quad	data 50%	3	15000	0.0773	0.0321	0.0247	0.0259
MNIST	FCN	learnt	f_quad	data 50%	4	15000	0.0808	0.0305	0.0230	0.0246
MNIST	FCN	learnt	f_quad	headline	0	30000	0.0572	0.0216	0.0170	0.0191
MNIST	FCN	learnt	f_quad	headline	1	30000	0.0574	0.0216	0.0178	0.0177
MNIST	FCN	learnt	f_quad	headline	2	30000	0.0572	0.0216	0.0178	0.0176
MNIST	FCN	learnt	f_quad	headline	3	30000	0.0595	0.0218	0.0178	0.0178
MNIST	FCN	learnt	f_quad	headline	4	30000	0.0589	0.0229	0.0176	0.0168
MNIST	FCN	learnt	f_quad	noise 10%	0	30000	0.2328	0.0626	0.0556	0.0393
MNIST	FCN	learnt	f_quad	noise 10%	1	30000	0.2335	0.0612	0.0563	0.0367
MNIST	FCN	learnt	f_quad	noise 10%	2	30000	0.2356	0.0589	0.0562	0.0379
MNIST	FCN	learnt	f_quad	noise 10%	3	30000	0.2313	0.0586	0.0527	0.0378
MNIST	FCN	learnt	f_quad	noise 10%	4	30000	0.2337	0.0629	0.0556	0.0386
MNIST	FCN	learnt	f_quad	noise 20%	0	30000	0.4246	0.1112	0.1095	0.0683
MNIST	FCN	learnt	f_quad	noise 20%	1	30000	0.4230	0.1196	0.1150	0.0685
MNIST	FCN	learnt	f_quad	noise 20%	2	30000	0.4369	0.1185	0.1150	0.0719
MNIST	FCN	learnt	f_quad	noise 20%	3	30000	0.4296	0.1164	0.1118	0.0681
MNIST	FCN	learnt	f_quad	noise 20%	4	30000	0.4305	0.1168	0.1045	0.0660
MNIST	FCN	learnt	f_quad	noise 40%	0	30000	0.6980	0.2649	0.2568	0.1814
MNIST	FCN	learnt	f_quad	noise 40%	1	30000	0.7015	0.2690	0.2582	0.1846
MNIST	FCN	learnt	f_quad	noise 40%	2	30000	0.6989	0.2641	0.2629	0.1813

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Dataset	Model	Prior	Obj.	Condition	Seed	n_b	Cert	Stoch	PM	Ens
MNIST	FCN	learnt	f_quad	noise 40%	3	30000	0.7048	0.2636	0.2481	0.1761
MNIST	FCN	learnt	f_quad	noise 40%	4	30000	0.7034	0.2619	0.2492	0.1746
MNIST	FCN	data-free	f_quad	headline	0	60000	0.3506	0.0974	0.0562	0.0604
MNIST	FCN	data-free	f_quad	headline	1	60000	0.3458	0.0922	0.0595	0.0569
MNIST	FCN	data-free	f_quad	headline	2	60000	0.3496	0.0914	0.0559	0.0568
MNIST	FCN	data-free	f_quad	headline	3	60000	0.3485	0.0884	0.0550	0.0569
MNIST	FCN	data-free	f_quad	headline	4	60000	0.3536	0.0941	0.0582	0.0612

B The unit-test suite

Table 7 lists the suite. Each row is one correction of Section 3.4 or one condition of Section 2, the number of tests defending it, and the fixed input and asserted output of each.

Table 7. The forty-two unit tests, grouped by what each defends. Every entry states the fixed input and the asserted output.

Correction	Tests	What is asserted
Bound-set sample count	4	A learnt-prior split of the full MNIST training set gives $n_{\text{post}} = 60{,}000$, $n_{\text{b}} = 30{,}000$, $n_{\text{prior}} = 30{,}000$; at 10% of the data, $6{,}000$ and $3{,}000$ exactly; a data-free prior gives $n_{\text{prior}} = 0$ and $n_{\text{b}} = \lvert S \rvert$.
Label-independent partition	6	$S_0 \cap S_{\text{b}} = \varnothing$ and $S_0 \cup S_{\text{b}} = S$; permuting every label leaves the partition bit-identical; the selected subset is spread across the index range rather than taken as a prefix; a different seed gives a different partition and the same seed reproduces it; and requesting a class-stratified partition for a learnt prior raises rather than being silently ignored.
Bound loader and loss range	6	The three bound-set loaders contain exactly S_{b} and no example of S_0, in one batch and when forced into several, while the posterior loader contains all of S; driving the production clamp and rescaling, the bounded cross-entropy lands in $[0, 1]$ at both extremes and over 200 random logit draws, and exceeds 1 on the same input with clamping off; and the runtime assertion that the loaders carry the saved partition fires when the evaluator is swapped onto a different subset, when a sampler covers only half of S_{b}, and when the trailing partial batch is dropped.
Monte-Carlo aggregation	2	With a deterministic posterior, evaluating in one batch of 130 and in batches of 32 with an unequal final batch agrees to 10^{-9} on the 0–1 estimate and 10^{-6} on the cross-entropy. With a stochastic posterior at $m = 200$ under a fixed seed the two batchings agree to 0.02 and 0.05.
Joint confidence	3	$\delta_{\text{MC}} = \delta_{\text{PAC}} = 0.005$ gives a joint confidence of 0.99 to 10^{-12}, and not $1 - \delta$; lowering δ_{PAC} raises the confidence and does not lower the certificate; kl^{-1} is monotone in the slack and never returns less than the empirical risk.

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Correction	Tests	What is asserted
Prior-selection union	5	At $K = 1$ the correction reproduces the as-run certificate exactly; it is monotone in K and never tightens; it leaves the Monte-Carlo inversion untouched; its cost is exactly log K in the numerator of the PAC-Bayes slack, not more, at $K \in \{2, 4, 16\}$; and the configured grid is the data-independent one, so reverting it to the values we happened to try fails here.
Pipeline wiring	8	Driving the real aggregation over a temporary results tree: the union is applied rather than merely imported, and published with and without; a learnt-prior run carrying no S_0 diagnostic is excluded rather than admitted, as is one whose prior sits at chance on S_0, and one whose stored certificate disagrees with its stored ingredients; a de-duplicated run is registered as superseded rather than dropped silently; the reseed occurs between the certificate and the deployment metrics; and re-drawing at two Monte-Carlo budgets leaves the deployment draw identical with the reseed and different without.
Seed control	4	The five streams derived from one base seed are pairwise distinct, all change when the base seed changes, the derivation is deterministic across processes, and the same base seed reproduces the whole bundle.
Tensor cache	2	The cached dataset returns tensors and labels bit-identical to the un-cached one, and the label-noise wrapper composes with it unchanged, so the speed-up cannot alter a reported number.
Result schema	2	A result round-trips through serialisation with n_{b} preserved, and the six distinct error fields are all present and separately named.

C Alternatives considered

This appendix records what was rejected in each of the design choices summarised in Section 3.3.

Which bound to report. The classical PAC-Bayes statements form a hierarchy, from the loose Hoeffding form through the tunable Catoni bound to the PAC-Bayes-kl inequality, the tightest but invertible only numerically. Since the question is how tight a certificate the method can deliver, a looser reporting bound would have flattened the differences between the objectives, so PAC-Bayes-kl is used throughout and the training objectives are treated only as different ways of producing the posterior that is then certified.

How to make a learnt prior admissible. A learnt prior is admissible only if it is independent of the data used to evaluate the bound, and there are two routes to that. The differential-privacy route of Dziugaite and Roy [5] lets the prior see all of the data at the price of an extra privacy term; the disjoint-subset route, which the reference uses and we follow, reserves a held-out subset the prior never trains on. We took the latter because it adds no extra slack and its independence condition can be checked directly, which suits an audit. Its cost is that the bound is then evaluated on fewer examples, a trade-off Section 4.3 returns to when interpreting the prior comparison.

The posterior family. A full-covariance Gaussian would carry a quadratic parameter count and a KL that does not scale to these networks. A Laplace posterior is equally admissible and tractable, but is not the reference's default, and changing it would have confounded the reproduction.

Spending the compute differently. Running fewer seeds at full Monte-Carlo budget was the obvious alternative to running five seeds at a reduced one. We rejected it because across-seed variability is one of the things the seed-control correction exists to expose: with the original pipeline every seed produced an identical run, so a study that could not measure spread would have left that correction untested.

D Reproduction

The artefact accompanying this dissertation is deposited at https://github.com/marcycn/pacbayes-cert; the version reported here is the one tagged there. Its layout is src/pacbayes_cert/ for the package, configs/ for one YAML file per headline cell named <dataset>_<objective>_<prior>_<model>.yaml, tests/ for the suite of Table 7, scripts/ for the orchestration, aggregation, figure and table steps, results/raw/ for one directory per run holding its resolved configuration, split indices, metrics and environment record, results/processed/ for the registry these are aggregated into, results/reference/ for the transcribed published values of Table 4, third_party/ for the vendored reference implementation, and legacy_invalidated/ for the pre-correction outputs. A single run, the full matrix, the tests and the derived outputs are reproduced by

```
python run.py --config configs/mnist_fquad_learnt_fcn.yaml --base-seed 0
SEEDS="0 1 2 3 4" bash scripts/run_matrix.sh
```

```
pytest tests/
python scripts/prior_fit_on_s0.py && python scripts/verify_splits.py
python scripts/aggregate.py && python scripts/gen_tables.py && python scripts/make_figures.py
python scripts/mc_sensitivity.py --run-dir results/raw/mnist_fquad_learnt_fcn_seed0 --with-analytic
python scripts/gibbs_per_class.py --run-dir results/raw/fashion-mnist_fquad_learnt_fcn_seed0
```

in that order, in an environment installed from the supplied lock file. The order matters at one point: `aggregate.py` applies the convergence screen of Section 3.8 to the prior network's error on S_0, so a learnt-prior run for which `prior_fit_on_s0.py` has not yet supplied that figure is registered as `unscreened` and excluded, rather than admitted on a default. Runs produced by the current `run.py` record it themselves and do not need that step; it exists to bring earlier runs onto the same footing. Each experiment is run from a fixed configuration file and a base seed; the processed registry, and from it every results table, figure and inline number in this dissertation, is regenerated from the stored per-run results, so no reported number is entered by hand. The forty-two unit tests behind the corrections of Section 3.4 and the conditions of Section 2 are listed in Table 7 and run from the supplied suite without a GPU. The library versions, hardware and software version of every run are recorded alongside its results, and the environment is pinned by the supplied lock file.

E Ethics and risk assessment

The project was assessed against the institution's ethics criteria before any experiment was run, and the determination was that no ethics application is required. The grounds are that it involves no human participants, no human or personal data, no external organisations and no primary data collection: the only data used are the three public benchmark datasets MNIST, Fashion-MNIST and CIFAR-10, obtained through their standard distributions and used in accordance with their published terms.

The certificate applies only to the Gibbs classifier, at the stated confidence, on the distribution from which the bound set is drawn; it does not certify a deployed deterministic network or performance under distribution shift (Sections 2.3 and 4.7). And a certificate is an aggregate over classes: Section 4.8 shows the error concentrating in a few confusable classes, so a single certified number can coexist with markedly unequal per-class performance, which matters wherever those classes correspond to groups a decision affects differently.

On physical risk, compute ran on a single desktop-class GPU; no hazardous equipment, no sensitive data and no fieldwork were involved.